

GCSE Physics

Exam Walkthrough Pack

Step-by-step worked exam-style questions with full explanations

Original JDSscience exam-style questions — not copied from past papers

Introduction

This Physics Exam Walkthrough Pack is designed to help you understand how to approach GCSE and IGCSE exam-style questions. Every question in this booklet is an original JDScience item written to match common specification skills — it is not copied from AQA, Edexcel, OCR or other past papers.

How to use this pack: read each question, cover the walkthrough, and try answering in exam conditions first. Then read what the question is asking, follow the step-by-step thinking, compare your answer with the model answer, and study the mark breakdown to see exactly how marks are awarded.

Command words tell you what type of answer is required. Section 1 explains the most common GCSE science command words. Always underline the command word before you start writing.

Showing working matters: in calculations, examiners award method marks even when the final answer is wrong. In longer answers, a clear logical sequence helps you secure every available mark.

Examiners award marks for correct science linked to the question. Keywords from the specification matter, but a correct idea in your own words can still score. Extended questions often use level-based marking: plan before you write, use paragraphs, and link ideas.

To move from Grade 4/5 to Grade 7/8/9: use precise terminology, explain mechanisms (not just names), include data or examples where asked, and check units and significant figures in calculations. For six-mark questions, a short plan and linked paragraphs beat a long unordered list.

Section 1: Command word guide

State

What you must do: Give a short, factual answer — often one word, number or phrase. No explanation is required unless the question also asks you to explain.

Common mistake: Writing a long paragraph when a single fact is enough, or giving an explanation when only a fact was asked for.

Model sentence starter: The [quantity/name] is ...

Give

What you must do: Provide the information requested. Similar to state, but may need a short phrase or named example.

Common mistake: Listing unrelated facts that do not answer the question.

Model sentence starter: One example is ... / The value is ...

Describe

What you must do: Say what happens or what something is like. Focus on observations, patterns or features. You do not need to say why.

Common mistake: Explaining causes when the command word is only describe.

Model sentence starter: First ... then ... / The pattern shows ...

Explain

What you must do: Give reasons why something happens. Link cause and effect using science ideas and key terms.

Common mistake: Describing what happens without saying why, or using everyday language instead of science.

Model sentence starter: This happens because ... which means ...

Calculate

What you must do: Work out a numerical answer. Show your working, include units, and give the final value to the required precision.

Common mistake: Missing units, no working, or using the wrong equation.

Model sentence starter: Use ... = ... / Substitute: ...

Compare

What you must do: Identify similarities and/or differences between two or more things. Refer to both sides.

Common mistake: Only describing one item, or listing features without comparing.

Model sentence starter: Both ... however ... / X has ... whereas Y ...

Evaluate

What you must do: Weigh up evidence or options, consider strengths and weaknesses, and reach a supported judgement.

Common mistake: Only listing advantages with no conclusion or balanced comment.

Model sentence starter: An advantage is ... A limitation is ... Overall ...

Suggest

What you must do: Apply your knowledge to a new situation. Your answer should be sensible and scientifically plausible.

Common mistake: Repeating textbook facts that do not fit the context given.

Model sentence starter: A possible reason is ... / This could be because ...

Justify

What you must do: Give evidence or reasons that support a choice, conclusion or statement already made.

Common mistake: Restating the claim without giving supporting science.

Model sentence starter: This is supported by ... because ...

Determine

What you must do: Find out a value or conclusion using data, a graph or a calculation. Show how you reached your answer.

Common mistake: Reading a graph incorrectly or not showing how the value was obtained.

Model sentence starter: From the graph ... / Using the data ...

Analyse

What you must do: Break information into parts, interpret patterns or trends, and explain what the data shows.

Common mistake: Describing data without interpreting it, or ignoring anomalies.

Model sentence starter: The data shows that ... / As X increases, Y ...

Section 2: Energy

QUESTION 1

State the unit of energy.

QUESTION TYPE

Topic: Energy | Command word: state | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

The question tests whether you know the SI unit used for energy.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The unit of energy is the joule (J).

MARK BREAKDOWN

Mark 1: Correct unit named as joule or J.

COMMON MISTAKES

- Writing watts (W) instead of joules.
- Spelling 'joules' without the symbol when asked for the unit.

EXAMINER TIP

Energy is measured in joules; power is measured in watts.

QUESTION 2

Give one example of a renewable energy resource.

QUESTION TYPE

Topic: Energy | Command word: give | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

You must name one energy resource that is naturally replenished.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

One example is solar energy.

MARK BREAKDOWN

Mark 1: Any valid renewable named, e.g. solar, wind, hydroelectric, tidal, geothermal, or biomass.

COMMON MISTAKES

- Naming a fossil fuel such as coal.
- Naming nuclear fuel without noting it is non-renewable.

EXAMINER TIP

Learn a short list of renewables and one non-renewable for quick comparison questions.

QUESTION 3

State what is meant by a closed system in energy transfers.

QUESTION TYPE

Topic: Energy | **Command word:** state | **Marks:** 1 | **Skill tested:** Knowledge recall

WHAT THE QUESTION IS ASKING

Define the idea of a closed system in the context of energy.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A closed system is one where no energy is transferred to or from the surroundings.

MARK BREAKDOWN

Mark 1: States that no energy enters or leaves the system.

COMMON MISTAKES

- Saying no matter can enter or leave (that describes a closed system in chemistry, not the energy definition needed here).
- Confusing closed with isolated systems at A-level.

EXAMINER TIP

In GCSE physics, closed system usually means the total energy stays the same.

QUESTION 4

Describe the energy transfer when a student lifts a book from the floor to a shelf.

QUESTION TYPE

Topic: Energy | **Command word:** describe | **Marks:** 2 | **Skill tested:** Energy transfers

WHAT THE QUESTION IS ASKING

Describe what happens to energy during this transfer without explaining why in detail.

STEP-BY-STEP THINKING

1. Identify the initial and final energy stores.
2. Name the method of transfer (work done by a force).
3. State energy gained by one store and supplied from another.
4. Use correct energy store names.
5. Check both stores are mentioned.

MODEL ANSWER

Chemical energy in the student's muscles decreases and gravitational potential energy of the book increases. Energy is transferred mechanically by the upward force doing work on the book.

MARK BREAKDOWN

Mark 1: Chemical store of muscles identified as decreasing.

Mark 2: Gravitational potential energy of book identified as increasing.

COMMON MISTAKES

- Saying energy is created.
- Using 'kinetic' only without mentioning the work done against gravity.

EXAMINER TIP

Always name both the store that decreases and the store that increases.

QUESTION 5

Explain why a car's brakes get hot when the car slows down.

QUESTION TYPE

Topic: Energy | **Command word:** explain | **Marks:** 2 | **Skill tested:** Energy dissipation

WHAT THE QUESTION IS ASKING

Give the reason linked to energy transfer when braking.

STEP-BY-STEP THINKING

1. Identify the kinetic energy store involved.
2. State where energy is transferred to.
3. Link heating to friction.
4. Use the idea of wasted or dissipated energy.
5. Check the explanation answers why, not just what happens.

MODEL ANSWER

The brakes get hot because kinetic energy of the car is transferred to the thermal store of the brakes and surroundings through friction, which dissipates useful energy as heat.

MARK BREAKDOWN

Mark 1: Kinetic energy decreases or is transferred.

Mark 2: Thermal energy increases in brakes due to friction / dissipation.

COMMON MISTAKES

- Saying energy is destroyed.
- Only stating brakes touch without mentioning energy stores.

EXAMINER TIP

Use 'transferred to the thermal store' rather than 'lost'.

QUESTION 6

Compare the energy changes in a pendulum at the highest point and at the bottom of its swing.

QUESTION TYPE

Topic: Energy | **Command word:** compare | **Marks:** 2 | **Skill tested:** Energy stores

WHAT THE QUESTION IS ASKING

Identify similarities and differences in energy stores at two positions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

At the highest point the pendulum has maximum gravitational potential energy and minimum kinetic energy. At the bottom, gravitational potential energy is minimum and kinetic energy is maximum. In both positions, if air resistance is ignored, total energy is conserved.

MARK BREAKDOWN

Mark 1: Correct stores compared at both positions (GPE high/low, KE low/high).

Mark 2: Similarity or conservation comment, or clear contrast of both positions.

COMMON MISTAKES

- Only describing one position.
- Ignoring kinetic energy at the bottom.

EXAMINER TIP

Compare command words need features of both situations.

QUESTION 7

Explain why insulating a house reduces heating costs.

QUESTION TYPE

Topic: Energy | Command word: explain | Marks: 3 | Skill tested: Efficiency and insulation

WHAT THE QUESTION IS ASKING

Link insulation to reduced energy transfer and cost.

STEP-BY-STEP THINKING

1. State what insulation reduces (thermal transfer).
2. Explain less energy leaves the home.
3. Link to heater switching on less often.
4. Connect to lower fuel/electrical use.
5. Use correct terminology: thermal store, transfer, efficiency.

MODEL ANSWER

Insulation reduces the rate of energy transfer by heating to the surroundings. Less energy must be supplied by the heating system to maintain the same temperature, so less fuel or electricity is used and heating costs fall.

MARK BREAKDOWN

Mark 1: Insulation reduces energy transfer to surroundings.

Mark 2: Less energy input needed to maintain temperature.

Mark 3: Lower energy use reduces cost.

COMMON MISTAKES

- Saying insulation creates energy.
- Only stating 'keeps heat in' without energy transfer language.

EXAMINER TIP

Link every efficiency answer to reduced wasted energy transfer.

QUESTION 8

A kettle transfers 180 kJ of energy to water in 3 minutes. Describe how you would calculate its power and state what extra information you already have.

QUESTION TYPE

Topic: Energy | Command word: describe | Marks: 3 | Skill tested: Power

WHAT THE QUESTION IS ASKING

Outline the calculation method using given data.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.

5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Power is calculated using $P = E \div t$. The energy transferred E is 180 kJ and the time t is 3 minutes, which must be converted to seconds (180 s) before substituting. Dividing energy by time gives power in watts if SI units are used.

MARK BREAKDOWN

Mark 1: States $P = E \div t$ or equivalent.

Mark 2: Identifies E and t from the question.

Mark 3: Notes time must be converted to seconds for SI power in watts.

COMMON MISTAKES

- Using minutes directly without conversion.
- Confusing power with energy.

EXAMINER TIP

Always convert minutes to seconds before calculating power in watts.

QUESTION 9

Evaluate the use of wind turbines compared with gas-fired power stations for generating electricity.

QUESTION TYPE

Topic: Energy | Command word: evaluate | Marks: 4 | Skill tested: Energy resources

WHAT THE QUESTION IS ASKING

Weigh advantages and disadvantages and reach a balanced judgement.

STEP-BY-STEP THINKING

1. Give advantage(s) of wind.
2. Give limitation(s) of wind.
3. Give advantage(s) of gas.
4. Give limitation(s) of gas.
5. Reach an overall judgement.

MODEL ANSWER

Wind turbines produce no direct carbon dioxide during operation and use a renewable resource, but output varies with weather and turbines need suitable locations. Gas-fired stations provide reliable, controllable output but burn fossil fuel, emit carbon dioxide, and contribute to climate change. Wind is better for lowering emissions long term, but gas currently helps meet demand when wind is low.

MARK BREAKDOWN

Mark 1: Valid advantage of wind.

Mark 2: Valid limitation of wind.

Mark 3: Valid advantage of gas.

Mark 4: Valid limitation of gas or overall judgement.

COMMON MISTAKES

- One-sided answer.
- No reference to reliability or emissions.

EXAMINER TIP

Evaluation needs points on both sides plus a conclusion.

QUESTION 10

Describe, with examples, the difference between renewable and non-renewable energy resources.

QUESTION TYPE

Topic: Energy | **Command word:** describe | **Marks:** 4 | **Skill tested:** Energy resources

WHAT THE QUESTION IS ASKING

Describe both categories and give an example of each.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Renewable resources are replenished as they are used, such as wind or solar. Non-renewable resources exist in finite amounts and will run out, such as coal, oil, and natural gas. Nuclear fuel is also treated as non-renewable at GCSE because uranium reserves are limited.

MARK BREAKDOWN

- Mark 1: Renewable definition with example.
- Mark 2: Non-renewable definition with example.
- Mark 3: Clear contrast between replenished and finite.
- Mark 4: Second valid point such as second example or time scale.

COMMON MISTAKES

- No examples.
- Calling nuclear renewable without qualification.

EXAMINER TIP

Pair each definition with a named example.

QUESTION 11

A motor lifts a 12 kg load through 2.5 m in 4.0 s. Calculate the power output of the motor. Use $g = 9.8 \text{ N/kg}$. Show your working.

QUESTION TYPE

Topic: Energy | **Command word:** calculate | **Marks:** 6 | **Skill tested:** Extended calculation

WHAT THE QUESTION IS ASKING

Calculate power from a lifting scenario using energy or work done.

STEP-BY-STEP THINKING

1. Calculate weight or force needed: $W = mg$.
2. Calculate work done or energy transferred: $E = F \times s$.
3. Convert time to seconds if needed.
4. Use $P = E \div t$.
5. Include units on the final answer.

MODEL ANSWER

Weight = $12 \times 9.8 = 117.6 \text{ N}$. Energy transferred = $117.6 \times 2.5 = 294 \text{ J}$. Power = $294 \div 4.0 = 73.5 \text{ W}$. The power output is 73 W (2 s.f.) or 74 W.

MARK BREAKDOWN

- Mark 1: Correct method for work/energy gained (294 J or equivalent).

Mark 2: Correct use of $P = E/t$.

Mark 3: Correct substitution.

Mark 4: Correct final value with unit W.

Mark 5: Working shown.

Mark 6: Appropriate significant figures.

COMMON MISTAKES

- Using mass instead of weight in $E_p = mgh$ without g.
- Forgetting units.
- Dividing by time in minutes.

EXAMINER TIP

For lifting at constant speed, work done on load equals energy transferred.

GRADE 7–9 EXTENSION

Alternatively, $P = Fv$ if speed is known: $v = 2.5/4.0 = 0.625$ m/s, $P = 117.6 \times 0.625 = 73.5$ W.

QUESTION 12

A 0.50 kg ball moving at 8.0 m/s is caught by a player. Calculate the kinetic energy of the ball.

QUESTION TYPE

Topic: Energy | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Calculate kinetic energy using mass and speed.

STEP-BY-STEP THINKING

1. Write $E_k = \frac{1}{2}mv^2$.
2. Substitute $m = 0.50$ kg and $v = 8.0$ m/s.
3. Square the speed before multiplying.
4. Include unit J on the answer.

MODEL ANSWER

$E_k = \frac{1}{2} \times 0.50 \times 8.0^2 = 0.25 \times 64 = 16$ J.

MARK BREAKDOWN

Mark 1: Correct equation $E_k = \frac{1}{2}mv^2$.

Mark 2: Correct substitution.

Mark 3: Correct answer 16 J with unit.

COMMON MISTAKES

- Forgetting to square velocity.
- Missing $\frac{1}{2}$ factor.
- No unit.

EXAMINER TIP

Always square the speed before multiplying by mass.

QUESTION 13

Describe how you would investigate the specific heat capacity of a metal block using an immersion heater, thermometer, and balance.

QUESTION TYPE

Topic: Energy | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe the method for measuring energy input and temperature change.

STEP-BY-STEP THINKING

1. Measure mass of block.
2. Measure initial temperature.
3. Supply known energy with heater for set time at known power.
4. Measure final temperature and calculate ΔT .
5. Reference $E = Pt$ and $E = mc\Delta T$.

MODEL ANSWER

Find the mass of the block on a balance. Measure initial temperature. Record the power of the immersion heater and time it is switched on so energy supplied $E = Pt$ is known. Measure final temperature and calculate temperature change ΔT . Use $c = E \div (m\Delta T)$ to find specific heat capacity. Repeat and calculate a mean.

MARK BREAKDOWN

- Mark 1: Mass measured.
- Mark 2: Initial and final temperatures measured.
- Mark 3: Energy input determined using $E = Pt$.
- Mark 4: Link to $c = E/m\Delta T$ or equivalent method steps.

COMMON MISTAKES

- Not explaining how energy input is found.
- Measuring only one temperature.

EXAMINER TIP

Practical descriptions must include how key quantities are measured.

Section 2: Electricity

QUESTION 14

State the relationship between current, charge, and time.

QUESTION TYPE

Topic: Electricity | Command word: state | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

Recall the equation linking charge, current, and time.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Charge flow $Q = I \times t$.

MARK BREAKDOWN

Mark 1: Correct relationship $Q = It$ stated.

COMMON MISTAKES

- Writing $V = IR$ instead.
- Reversing the relationship to $t = Q/I$ when only the relationship is asked.

EXAMINER TIP

Say 'charge flow equals current times time'.

QUESTION 15

Give the unit of electrical resistance.

QUESTION TYPE

Topic: Electricity | Command word: give | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

Name the unit for resistance.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The unit of resistance is the ohm (Ω).

MARK BREAKDOWN

Mark 1: Ohm or Ω .

COMMON MISTAKES

- Writing volts or amps.
- Spelling ohm incorrectly without symbol.

EXAMINER TIP

Resistance uses Ω ; current uses A; potential difference uses V.

QUESTION 16

State what happens to the current in a series circuit if one lamp is removed and the circuit breaks.

QUESTION TYPE

Topic: Electricity | **Command word:** state | **Marks:** 1 | **Skill tested:** Series circuits

WHAT THE QUESTION IS ASKING

State the effect on current when the circuit is open.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The current stops flowing / becomes zero because the circuit is incomplete.

MARK BREAKDOWN

Mark 1: Current stops or becomes zero due to broken/open circuit.

COMMON MISTAKES

- Saying current increases.
- Saying other lamps get brighter in a broken circuit.

EXAMINER TIP

Open circuit means no complete path for charge flow.

QUESTION 17

Describe the difference between series and parallel circuits for lamp brightness when another lamp is added.

QUESTION TYPE

Topic: Electricity | **Command word:** describe | **Marks:** 2 | **Skill tested:** Circuit arrangements

WHAT THE QUESTION IS ASKING

Compare how adding a lamp affects others in series and parallel.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

In series, adding another lamp increases total resistance so current decreases and all lamps become dimmer. In parallel, each lamp is across the full supply voltage so each stays equally bright when another branch is added.

MARK BREAKDOWN

Mark 1: Series: current/resistance effect makes lamps dimmer.

Mark 2: Parallel: each lamp has full p.d. so brightness unchanged.

COMMON MISTAKES

- Only describing one type.
- Saying parallel lamps dim when another is added.

EXAMINER TIP

Link brightness to current in series and p.d. in parallel.

QUESTION 18

Explain why metals are good conductors of electricity.

QUESTION TYPE

Topic: Electricity | Command word: explain | Marks: 2 | Skill tested: Conduction

WHAT THE QUESTION IS ASKING

Explain using charge carriers in metals.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Metals contain delocalised electrons that are free to move through the structure. When a potential difference is applied, these electrons drift through the metal, carrying charge and creating an electric current.

MARK BREAKDOWN

Mark 1: Delocalised/free electrons mentioned.

Mark 2: Electrons move when p.d. applied causing current.

COMMON MISTAKES

- Referring to protons moving.
- Saying atoms move through the wire.

EXAMINER TIP

Conductors: free electrons. Insulators: electrons cannot move freely.

QUESTION 19

A student connects a 6.0 V battery to a resistor of 12 Ω . Calculate the current in the resistor.

QUESTION TYPE

Topic: Electricity | Command word: calculate | Marks: 2 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Use $V = IR$ to find current.

STEP-BY-STEP THINKING

1. Rearrange to $I = V \div R$.
2. Substitute values.
3. Calculate and give unit A.

MODEL ANSWER

$$I = V \div R = 6.0 \div 12 = 0.50 \text{ A.}$$

MARK BREAKDOWN

Mark 1: Correct rearrangement $I = V/R$.

Mark 2: Correct answer 0.50 A with unit.

COMMON MISTAKES

- Multiplying V and R.
- Unit mA without conversion shown.

EXAMINER TIP

Show rearrangement before substituting.

QUESTION 20

Explain why electrical appliances with metal cases must be earthed.

QUESTION TYPE

Topic: Electricity | **Command word:** explain | **Marks:** 3 | **Skill tested:** Mains safety

WHAT THE QUESTION IS ASKING

Link earthing to safety if live wire touches the case.

STEP-BY-STEP THINKING

1. State case could become live if fault occurs.
2. Earth wire provides low-resistance path to ground.
3. Large current flows, fuse blows or breaker trips.
4. Prevents user receiving shock.

MODEL ANSWER

If the live wire touches the metal case, the case becomes dangerous. The earth wire connects the case to the ground so a large current flows to earth, which melts the fuse or trips the circuit breaker, disconnecting the supply and preventing a person from receiving a fatal shock.

MARK BREAKDOWN

- Mark 1: Fault makes case live.
Mark 2: Earth provides path to ground causing large current.
Mark 3: Fuse/breaker disconnects supply protecting user.

COMMON MISTAKES

- Saying earth prevents all current.
- No link to fuse or breaker.

EXAMINER TIP

Earth wire works with fuse or RCD to protect users.

QUESTION 21

Describe how the total resistance of two identical resistors changes when connected in series and when connected in parallel.

QUESTION TYPE

Topic: Electricity | **Command word:** describe | **Marks:** 3 | **Skill tested:** Combined resistance

WHAT THE QUESTION IS ASKING

Describe both combinations for identical resistors.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

In series, total resistance is the sum of individual resistances, so two identical resistors double the total resistance. In parallel, total resistance is less than either resistor alone because there are more paths for current; for two identical resistors, total resistance is half of one resistor.

MARK BREAKDOWN

Mark 1: Series: resistances add, total doubles for identical pair.

Mark 2: Parallel: total is less; half value for two identical.

Mark 3: Both series and parallel described.

COMMON MISTAKES

- Saying parallel resistance is larger.
- Only one arrangement described.

EXAMINER TIP

Series adds; parallel reduces total resistance.

QUESTION 22

A filament lamp has a resistance of $120\ \Omega$ at normal brightness. Calculate the charge that flows through the lamp in 2 minutes when connected to 12 V. Show your working.

QUESTION TYPE

Topic: Electricity | Command word: calculate | Marks: 4 | Skill tested: Extended calculation

WHAT THE QUESTION IS ASKING

Combine $V = IR$ and $Q = It$.

STEP-BY-STEP THINKING

1. Find current using $I = V/R$.
2. Convert time to seconds.
3. Use $Q = It$.
4. Include unit C.

MODEL ANSWER

$I = 12 \div 120 = 0.10\ \text{A}$. $t = 2 \times 60 = 120\ \text{s}$. $Q = 0.10 \times 120 = 12\ \text{C}$.

MARK BREAKDOWN

Mark 1: Current calculated correctly.

Mark 2: Time converted to seconds.

Mark 3: $Q = It$ used correctly.

Mark 4: Answer 12 C with unit.

COMMON MISTAKES

- Using minutes in $Q = It$.
- Wrong current from V/R error.

EXAMINER TIP

Multi-step electricity questions: solve one unknown at a time.

QUESTION 23

Compare direct current (d.c.) and alternating current (a.c.).

QUESTION TYPE

Topic: Electricity | **Command word:** compare | **Marks:** 4 | **Skill tested:** Current types

WHAT THE QUESTION IS ASKING

Identify differences between d.c. and a.c.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Direct current flows in one direction only, such as from a cell. Alternating current repeatedly reverses direction, as supplied by the UK mains at about 50 Hz. Mains a.c. is also at much higher voltage than typical cells.

MARK BREAKDOWN

- Mark 1: d.c. one direction.
- Mark 2: a.c. reverses direction.
- Mark 3: Example or frequency/mains context.
- Mark 4: Additional valid difference such as source or voltage.

COMMON MISTAKES

- Saying a.c. does not change direction.
- No comparison of both.

EXAMINER TIP

UK mains: 230 V a.c. at 50 Hz.

QUESTION 24

The National Grid uses step-up transformers at power stations and step-down transformers near homes. Explain why transformers are used in this way, including how they affect current in the transmission cables and energy loss.

QUESTION TYPE

Topic: Electricity | **Command word:** explain | **Marks:** 6 | **Skill tested:** Power transmission

WHAT THE QUESTION IS ASKING

Explain step-up and step-down roles and link to reduced heating in cables.

STEP-BY-STEP THINKING

1. Step-up increases voltage for transmission.
2. For same power, higher voltage means lower current ($P = VI$).
3. Lower current reduces energy loss by heating in cables (I^2R).
4. Step-down reduces voltage to safe 230 V for homes.
5. Transformers only work with a.c.

MODEL ANSWER

Power stations use step-up transformers to increase transmission voltage, which reduces current in the cables for the same power transferred. Lower current reduces wasted energy as heat in the cables because power loss depends on I^2R . Near homes, step-down transformers reduce voltage to a safe 230 V. Transformers require alternating current to operate.

MARK BREAKDOWN

Mark 1: Step-up increases voltage for transmission.

Mark 2: Lower current for same power.

Mark 3: Reduced I^2R energy loss in cables.

Mark 4: Step-down for safe domestic voltage.

Mark 5: Transformers and a.c. link.

Mark 6: Clear logical explanation throughout.

COMMON MISTAKES

- Saying transformers increase power.
- No link to I^2R losses.

EXAMINER TIP

Higher transmission voltage → lower current → less wasted heat.

QUESTION 25

A charge of 450 C flows through a lamp in 3 minutes. Calculate the current in the lamp.

QUESTION TYPE

Topic: Electricity | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Calculate current from charge flow and time.

STEP-BY-STEP THINKING

1. Use $I = Q \div t$.
2. Convert 3 minutes to 180 s.
3. Divide and give answer in amps.

MODEL ANSWER

$t = 3 \times 60 = 180 \text{ s}$. $I = 450 \div 180 = 2.5 \text{ A}$.

MARK BREAKDOWN

Mark 1: Time converted to seconds.

Mark 2: Correct use of $I = Q/t$.

Mark 3: Answer 2.5 A with unit.

COMMON MISTAKES

- Using minutes directly.
- Inverting equation incorrectly.

EXAMINER TIP

Charge flow questions almost always need seconds.

QUESTION 26

Describe how you would investigate how the resistance of a wire depends on its length using a circuit with an ammeter, voltmeter, and metre rule.

QUESTION TYPE

Topic: Electricity | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe controlling variables and taking repeat readings.

STEP-BY-STEP THINKING

1. Set up wire, ammeter in series, voltmeter in parallel.
2. Measure length with metre rule.

3. Record current and p.d., calculate $R = V/I$.
4. Keep material, diameter, and temperature constant.
5. Repeat and plot R against length.

MODEL ANSWER

Connect the wire in series with ammeter and power supply; connect voltmeter in parallel with the wire. Measure length with a metre rule. Record current and potential difference, then calculate $R = V/I$. Keep material, thickness, and temperature constant while changing only length. Take repeat readings for each length, remove anomalies, and plot a graph of resistance against length.

MARK BREAKDOWN

- Mark 1: Correct circuit description.
- Mark 2: $R = V/I$ used.
- Mark 3: Length measured.
- Mark 4: Control variables and repeats mentioned.

COMMON MISTAKES

- Ammeter in parallel.
- No control variables.

EXAMINER TIP

Required practical answers should name equipment and control variables.

Section 2: Particle model of matter

QUESTION 27

State what is meant by density.

QUESTION TYPE

Topic: Particle model of matter | Command word: state | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

Define density as a property of matter.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Density is mass per unit volume.

MARK BREAKDOWN

Mark 1: Mass per unit volume stated.

COMMON MISTAKES

- Saying weight per volume.
- No reference to volume.

EXAMINER TIP

Use 'mass per unit volume' exactly.

QUESTION 28

Give the state of matter in which particles are close together but can slide past each other.

QUESTION TYPE

Topic: Particle model of matter | Command word: give | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

Name the state with fixed volume but changeable shape.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Liquid.

MARK BREAKDOWN

Mark 1: Liquid identified.

COMMON MISTAKES

- Gas.
- Solid when sliding is mentioned.

EXAMINER TIP

Solid fixed shape; liquid fixed volume; gas fills container.

QUESTION 29

State what happens to the pressure of a fixed mass of gas when its volume is decreased at constant temperature.

QUESTION TYPE

Topic: Particle model of matter | **Command word:** state | **Marks:** 1 | **Skill tested:** Gas pressure

WHAT THE QUESTION IS ASKING

State the effect on pressure when volume decreases.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Pressure increases.

MARK BREAKDOWN

Mark 1: Pressure increases.

COMMON MISTAKES

- Pressure decreases.
- Saying temperature must rise without question saying so.

EXAMINER TIP

Boyle's law: smaller volume → higher pressure at constant temperature.

QUESTION 30

Describe the arrangement and movement of particles in a solid.

QUESTION TYPE

Topic: Particle model of matter | **Command word:** describe | **Marks:** 2 | **Skill tested:** Particle model

WHAT THE QUESTION IS ASKING

Describe particle positions and motion in a solid.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Particles in a solid are close together in a regular arrangement and vibrate about fixed positions.

MARK BREAKDOWN

Mark 1: Close together / regular arrangement.

Mark 2: Vibrate about fixed positions.

COMMON MISTAKES

- Saying particles move freely.
- No mention of vibration.

EXAMINER TIP

Solids: strong forces, fixed positions, vibrate.

QUESTION 31

Explain why a balloon expands when heated.

QUESTION TYPE

Topic: Particle model of matter | Command word: explain | Marks: 2 | Skill tested: Gas behaviour

WHAT THE QUESTION IS ASKING

Link heating to particle speed and pressure.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Heating increases the average kinetic energy of air particles so they move faster and hit the balloon walls more often and with greater force, increasing pressure and causing expansion.

MARK BREAKDOWN

Mark 1: Particles move faster / more kinetic energy.

Mark 2: More frequent or harder collisions increase pressure causing expansion.

COMMON MISTAKES

- Saying particles get bigger.
- No link to collisions.

EXAMINER TIP

Heating gases: faster particles → more pressure.

QUESTION 32

Compare the density of a solid block and the same material after it melts.

QUESTION TYPE

Topic: Particle model of matter | Command word: compare | Marks: 2 | Skill tested: Density

WHAT THE QUESTION IS ASKING

Compare density before and after melting for same material.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

When the solid melts to a liquid, particles move slightly further apart so volume increases slightly while mass stays the same, meaning density decreases.

MARK BREAKDOWN

Mark 1: Volume increases when melted.

Mark 2: Density decreases because mass unchanged.

COMMON MISTAKES

- Saying density increases.
- Only describing one state.

EXAMINER TIP

Same mass, larger volume → lower density.

QUESTION 33

A block has mass 540 g and volume 200 cm³. Calculate its density in g/cm³.

QUESTION TYPE

Topic: Particle model of matter | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Use density = mass ÷ volume.

STEP-BY-STEP THINKING

1. Write $\rho = m/V$.
2. Substitute 540 and 200.
3. Divide and include unit.
4. Check answer sensible.

MODEL ANSWER

$$\rho = 540 \div 200 = 2.7 \text{ g/cm}^3.$$

MARK BREAKDOWN

- Mark 1: Correct equation.
Mark 2: Correct substitution.
Mark 3: Answer 2.7 g/cm³.

COMMON MISTAKES

- Multiplying mass and volume.
- Unit kg/m³ without conversion.

EXAMINER TIP

Always state unit g/cm³ or kg/m³ clearly.

QUESTION 34

Explain why steam burns can be more severe than boiling water burns at 100 °C.

QUESTION TYPE

Topic: Particle model of matter | Command word: explain | Marks: 3 | Skill tested: Internal energy

WHAT THE QUESTION IS ASKING

Link latent heat of vaporisation to extra energy transfer.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Steam at 100 °C condenses on skin and releases latent heat of vaporisation as well as cooling from 100 °C, transferring more energy than water at 100 °C alone.

MARK BREAKDOWN

Mark 1: Steam condenses releasing latent heat.

Mark 2: Extra energy transferred beyond 100 °C water.

Mark 3: More severe burn linked to extra energy.

COMMON MISTAKES

- Only stating steam is hotter than 100 °C.
- No latent heat idea.

EXAMINER TIP

Condensation releases latent heat — a favourite exam point.

QUESTION 35

Describe how you would determine the density of an irregular stone using a balance, displacement can, and measuring cylinder.

QUESTION TYPE

Topic: Particle model of matter | **Command word:** describe | **Marks:** 4 | **Skill tested:** Required practical skills

WHAT THE QUESTION IS ASKING

Describe measuring mass and volume by displacement.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Find mass on a balance. Fill a displacement can to the spout, then submerge the stone and collect displaced water in a measuring cylinder to find volume. Calculate density using $\rho = m/V$. Repeat and mean.

MARK BREAKDOWN

Mark 1: Mass measured on balance.

Mark 2: Volume by displacement.

Mark 3: $\rho = m/V$ used.

Mark 4: Repeat/mean mentioned.

COMMON MISTAKES

- Measuring volume with a ruler.
- No displacement method.

EXAMINER TIP

Irregular objects need displacement for volume.

QUESTION 36

Evaluate using the particle model to explain why liquids fill the bottom of a container but gases fill all available space.

QUESTION TYPE

Topic: Particle model of matter | **Command word:** evaluate | **Marks:** 4 | **Skill tested:** Particle model

WHAT THE QUESTION IS ASKING

Use particle spacing and forces to evaluate the explanation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.

2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The model explains that liquid particles are close but mobile, so they stay together under gravity at the bottom, whereas gas particles are far apart with weak forces and move randomly in all directions, spreading throughout any volume. The model is useful but does not predict exact evaporation rates without more detail.

MARK BREAKDOWN

- Mark 1: Liquid particles close, move but stay together.
- Mark 2: Gas particles far apart, random motion fills space.
- Mark 3: Evaluation of model usefulness or limitation.
- Mark 4: Balanced conclusion.

COMMON MISTAKES

- No comparison of both states.
- No evaluation judgement.

EXAMINER TIP

Evaluate = explain plus judge how good the model is.

QUESTION 37

A 2.0 kg copper block is heated from 20 °C to 80 °C. The specific heat capacity of copper is 385 J/kg °C. Calculate the energy transferred to the block.

QUESTION TYPE

Topic: Particle model of matter | Command word: calculate | Marks: 6 | Skill tested: Extended calculation

WHAT THE QUESTION IS ASKING

Use $E = mc\Delta T$.

STEP-BY-STEP THINKING

1. Find $\Delta T = 80 - 20$.
2. Use $E = mc\Delta T$.
3. Substitute all values.
4. Calculate with unit J.
5. Show working.
6. Check sign and unit.

MODEL ANSWER

$\Delta T = 60\text{ °C}$. $E = 2.0 \times 385 \times 60 = 46\,200\text{ J} = 4.62 \times 10^4\text{ J}$.

MARK BREAKDOWN

- Mark 1: ΔT correct.
- Mark 2: $E = mc\Delta T$ used.
- Mark 3: Substitution correct.
- Mark 4: Answer 46 200 J.
- Mark 5: Unit J.
- Mark 6: Working shown.

COMMON MISTAKES

- Using 80 instead of ΔT .
- Missing unit.

EXAMINER TIP

Specific heat capacity needs temperature change not final temperature.

QUESTION 38

Calculate the density of a material if a sample of mass 1.2 kg has a volume of 0.00040 m³. Give your answer in kg/m³.

QUESTION TYPE

Topic: Particle model of matter | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Calculate density in SI units.

STEP-BY-STEP THINKING

1. Use $\rho = m/V$.
2. Substitute values.
3. Divide and give kg/m³.

MODEL ANSWER

$$\rho = 1.2 \div 0.00040 = 3000 \text{ kg/m}^3.$$

MARK BREAKDOWN

- Mark 1: Correct equation.
- Mark 2: Correct substitution.
- Mark 3: Answer 3000 kg/m³.

COMMON MISTAKES

- Incorrect power of ten.
- Unit g/cm³ only.

EXAMINER TIP

Convert cm³ to m³ carefully in mixed unit questions.

QUESTION 39

Describe how the particle model explains changes of state when ice melts to form water.

QUESTION TYPE

Topic: Particle model of matter | Command word: describe | Marks: 4 | Skill tested: Changes of state

WHAT THE QUESTION IS ASKING

Describe particle behaviour during melting without calculating energy values.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

In ice, particles vibrate in fixed positions in a regular lattice. When melting, energy increases until particles break free from fixed positions and move while staying close together, becoming liquid with greater disorder but similar spacing.

MARK BREAKDOWN

- Mark 1: Solid particles vibrate in fixed positions.
- Mark 2: Energy input during melting.
- Mark 3: Particles move but remain close in liquid.
- Mark 4: Link to increased disorder without large spacing change.

COMMON MISTAKES

- Saying particles separate completely like a gas.
- No mention of energy input.

EXAMINER TIP

Melting increases kinetic energy until bonds in the lattice break.

Section 2: Atomic structure and radioactivity

QUESTION 40

State the name of radiation consisting of helium nuclei.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: state | Marks: 1 | Skill tested: Radioactivity

WHAT THE QUESTION IS ASKING

Identify alpha radiation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Alpha (α) radiation.

MARK BREAKDOWN

Mark 1: Alpha or α .

COMMON MISTAKES

- Beta.
- Gamma.

EXAMINER TIP

α = 2 protons + 2 neutrons.

QUESTION 41

Give one source of background radiation.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: give | Marks: 1 | Skill tested: Background radiation

WHAT THE QUESTION IS ASKING

Name a natural or man-made background source.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Radon gas from rocks (or cosmic rays / medical sources / nuclear weapons fallout).

MARK BREAKDOWN

Mark 1: Any valid background source.

COMMON MISTAKES

- Microwave ovens (not ionising background).
- Visible light.

EXAMINER TIP

Radon from rocks is a common UK example.

QUESTION 42

State what is meant by the half-life of a radioactive isotope.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: state | Marks: 1 | Skill tested: Half-life

WHAT THE QUESTION IS ASKING

Define half-life.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Half-life is the time taken for the activity (or number of undecayed nuclei) to halve.

MARK BREAKDOWN

Mark 1: Time for activity or undecayed nuclei to halve.

COMMON MISTAKES

- Half the life of atom.
- Time for all nuclei to decay.

EXAMINER TIP

Half-life is statistical — not when all nuclei are gone.

QUESTION 43

Describe what happens to the nucleus during beta-minus decay.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: describe | Marks: 2 | Skill tested: Nuclear decay

WHAT THE QUESTION IS ASKING

Describe the change in the nucleus when beta is emitted.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A neutron changes into a proton and an electron; the electron is emitted as beta radiation and the nucleus has one more proton so atomic number increases by one.

MARK BREAKDOWN

Mark 1: Neutron becomes proton and electron.

Mark 2: Beta particle emitted; atomic number increases.

COMMON MISTAKES

- Proton becomes neutron without specifying beta-minus.
- Nucleus unchanged.

EXAMINER TIP

Beta-minus: neutron $\rightarrow$ proton + electron.

QUESTION 44

Explain why alpha radiation is more ionising than gamma radiation.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: explain | Marks: 2 | Skill tested: Radiation properties

WHAT THE QUESTION IS ASKING

Link particle size/charge to ionisation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Alpha particles are large and doubly charged so they strongly interact with atoms, removing electrons over a short distance. Gamma photons have less charge interaction per collision and are less ionising.

MARK BREAKDOWN

Mark 1: Alpha large/charged, strong interactions.

Mark 2: More ionisation per alpha particle than gamma.

COMMON MISTAKES

- Saying gamma has more energy so more ionising.
- No interaction argument.

EXAMINER TIP

Ionising power $\neq$ penetrating power.

QUESTION 45

Compare how alpha, beta, and gamma radiation are stopped.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: compare | Marks: 2 | Skill tested: Penetration

WHAT THE QUESTION IS ASKING

Compare stopping materials for three types.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Alpha is stopped by paper or a few centimetres of air. Beta is stopped by thin aluminium. Gamma requires thick lead or thick concrete to be significantly reduced.

MARK BREAKDOWN

Mark 1: Alpha stopped by paper/air.

Mark 2: Beta stopped by aluminium.

Mark 3: Gamma reduced by thick lead/concrete.

COMMON MISTAKES

- All stopped by paper.
- Only one type described.

EXAMINER TIP

Penetration increases $\alpha < \beta < \gamma$.

QUESTION 46

Explain why radioactive sources used in medical tracers must have a short half-life.

QUESTION TYPE

Topic: Atomic structure and radioactivity | **Command word:** explain | **Marks:** 3 | **Skill tested:** Uses of radiation

WHAT THE QUESTION IS ASKING

Link half-life to dose and safety.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A short half-life means activity falls quickly so the patient receives a lower total dose of radiation. The tracer remains detectable long enough for diagnosis then decays rapidly.

MARK BREAKDOWN

- Mark 1: Short half-life reduces dose.
Mark 2: Still detectable for scan.
Mark 3: Decays quickly after use.

COMMON MISTAKES

- Long half-life preferred.
- No dose argument.

EXAMINER TIP

Medical tracers: short half-life, gamma emitter.

QUESTION 47

A radioactive sample has activity 800 counts/min. After 2 half-lives, what is the activity?

QUESTION TYPE

Topic: Atomic structure and radioactivity | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Half-life calculation

WHAT THE QUESTION IS ASKING

Apply halving twice.

STEP-BY-STEP THINKING

1. Activity halves each half-life.
2. Apply twice.
3. State final value with unit.

MODEL ANSWER

After 1 half-life: 400 counts/min. After 2 half-lives: 200 counts/min.

MARK BREAKDOWN

- Mark 1: First halving to 400.

Mark 2: Second halving to 200.

Mark 3: Correct final value.

COMMON MISTAKES

- Dividing by 4 without showing steps.
- Adding instead of halving.

EXAMINER TIP

Each half-life multiplies activity by $\frac{1}{2}$.

QUESTION 48

Describe the plum pudding model and how the alpha scattering experiment led to the nuclear model.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: describe | Marks: 4 | Skill tested: Atomic models

WHAT THE QUESTION IS ASKING

Describe both models and evidence from Rutherford.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Plum pudding: positive charge spread throughout with electrons embedded. Rutherford fired alpha particles at thin gold foil; most passed through but some deflected through large angles. This showed most of the atom is empty space with a small, dense, positively charged nucleus.

MARK BREAKDOWN

Mark 1: Plum pudding described.

Mark 2: Most alphas passed through.

Mark 3: Large deflections showed small dense nucleus.

Mark 4: Nuclear model conclusion.

COMMON MISTAKES

- Electrons in nucleus.
- No link to deflection evidence.

EXAMINER TIP

Large angle scattering → small massive nucleus.

QUESTION 49

Evaluate the use of nuclear fission for generating electricity.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: evaluate | Marks: 4 | Skill tested: Nuclear power

WHAT THE QUESTION IS ASKING

Weigh benefits and risks.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.

3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fission produces large energy from small fuel and no direct CO₂ during operation, giving reliable baseload power. However, radioactive waste must be stored safely, accidents can release contamination, and plant decommissioning is costly. Useful low-carbon source if waste and safety are managed.

MARK BREAKDOWN

- Mark 1: High energy output/low CO₂.
- Mark 2: Waste or accident risk.
- Mark 3: Reliability advantage.
- Mark 4: Overall judgement.

COMMON MISTAKES

- Only advantages.
- Confusing fission with fusion.

EXAMINER TIP

Fission = large nuclei split; fusion = small nuclei join.

QUESTION 50

An isotope has a half-life of 6 hours. A sample has initial activity 320 Bq. Calculate the activity after 18 hours.

QUESTION TYPE

Topic: Atomic structure and radioactivity | **Command word:** calculate | **Marks:** 6 | **Skill tested:** Extended calculation

WHAT THE QUESTION IS ASKING

Determine number of half-lives and apply repeatedly.

STEP-BY-STEP THINKING

1. $18 \div 6 = 3$ half-lives.
2. Halve three times or multiply by $(\frac{1}{2})^3$.
3. Calculate final activity.
4. Include unit Bq.
5. Show working.
6. Check answer.

MODEL ANSWER

18 hours = 3 half-lives. Activity = $320 \times (\frac{1}{2})^3 = 320 \times 0.125 = 40$ Bq.

MARK BREAKDOWN

- Mark 1: 3 half-lives identified.
- Mark 2: Correct multiplier $(\frac{1}{2})^3$.
- Mark 3: Answer 40 Bq.
- Mark 4: Unit Bq.
- Mark 5: Working.
- Mark 6: Logical method.

COMMON MISTAKES

- Dividing by 3 instead of halving.
- Wrong number of half-lives.

EXAMINER TIP

Always find number of half-lives first.

QUESTION 51

Radon-222 has a half-life of 3.8 days. A sample contains 1200 undecayed nuclei. Estimate how many remain after 7.6 days.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Apply two half-lives.

STEP-BY-STEP THINKING

1. $7.6 \div 3.8 = 2$ half-lives.
2. Halve twice.
3. State integer nuclei.

MODEL ANSWER

2 half-lives: $1200 \rightarrow 600 \rightarrow 300$ nuclei remain.

MARK BREAKDOWN

- Mark 1: 2 half-lives.
Mark 2: First halving.
Mark 3: Answer 300 nuclei.

COMMON MISTAKES

- Linear subtraction.
- Wrong half-life count.

EXAMINER TIP

Half-life questions rarely need exact exponentials at GCSE.

QUESTION 52

Describe how you would investigate the thickness of paper required to absorb beta radiation from a source, safely.

QUESTION TYPE

Topic: Atomic structure and radioactivity | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe safe method with detector and absorbers.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Set source in lead collimator at fixed distance from G-M tube. Measure count rate with no absorber using timer. Place increasing sheets of paper, record count rate each time. Use tongs, keep distance, minimise time, plot count rate against number of sheets.

MARK BREAKDOWN

- Mark 1: Fixed distance and timing.
Mark 2: Variable absorber thickness.
Mark 3: Safety: distance, tongs, minimise time.
Mark 4: Repeat readings.

COMMON MISTAKES

- Handling source with bare hands.
- No control of distance.

EXAMINER TIP

Radiation practicals must include safety precautions.

Section 2: Forces

QUESTION 53

State the unit of force.

QUESTION TYPE

Topic: Forces | Command word: state | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

Name SI unit of force.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The newton (N).

MARK BREAKDOWN

Mark 1: Newton or N.

COMMON MISTAKES

- Joule.
- Kg.

EXAMINER TIP

Force N; mass kg; weight N.

QUESTION 54

Give the equation linking force, mass, and acceleration.

QUESTION TYPE

Topic: Forces | Command word: give | Marks: 1 | Skill tested: Equations

WHAT THE QUESTION IS ASKING

Recall Newton's second law equation.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

$F = ma$.

MARK BREAKDOWN

Mark 1: $F = ma$.

COMMON MISTAKES

- $F = mv$.
- $a = F/m$ reversed incorrectly as the equation.

EXAMINER TIP

$F = ma$ is in the physics equation sheet.

QUESTION 55

State what is meant by the resultant force on an object.

QUESTION TYPE

Topic: Forces | Command word: state | Marks: 1 | Skill tested: Resultant forces

WHAT THE QUESTION IS ASKING

Define resultant force.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The resultant force is the single force that has the same effect as all forces acting on the object combined.

MARK BREAKDOWN

Mark 1: Single force equivalent to combined forces.

COMMON MISTAKES

- Largest force only.
- Weight only.

EXAMINER TIP

Resultant decides acceleration direction.

QUESTION 56

Describe the motion of an object when the resultant force is zero.

QUESTION TYPE

Topic: Forces | Command word: describe | Marks: 2 | Skill tested: Newton's first law

WHAT THE QUESTION IS ASKING

Describe velocity when resultant force zero.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

If the object is stationary it remains at rest. If it is moving it continues at constant velocity in a straight line.

MARK BREAKDOWN

Mark 1: Stationary stays at rest.

Mark 2: Moving continues at constant velocity.

COMMON MISTAKES

- Object always stops.
- Speed always zero.

EXAMINER TIP

Zero resultant → no change in velocity.

QUESTION 57

Explain why a seat belt reduces injury in a car crash.

QUESTION TYPE

Topic: Forces | Command word: explain | Marks: 2 | Skill tested: Safety

WHAT THE QUESTION IS ASKING

Link increased stopping time to reduced force.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The seat belt increases the time taken for the passenger to stop. For the same change in momentum, a longer time reduces the rate of change of momentum, so the force on the passenger is smaller.

MARK BREAKDOWN

Mark 1: Increases stopping time.

Mark 2: Reduces force for same momentum change.

COMMON MISTAKES

- Saying belt reduces momentum.
- No time or force link.

EXAMINER TIP

$F = \Delta p / \Delta t$ — longer time $\rightarrow$ smaller force.

QUESTION 58

A car of mass 1200 kg accelerates at 2.5 m/s². Calculate the resultant force.

QUESTION TYPE

Topic: Forces | Command word: calculate | Marks: 2 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Use $F = ma$.

STEP-BY-STEP THINKING

1. Write $F = ma$.
2. Substitute.
3. Answer in N.

MODEL ANSWER

$F = 1200 \times 2.5 = 3000 \text{ N}$.

MARK BREAKDOWN

Mark 1: $F = ma$.

Mark 2: Answer 3000 N.

COMMON MISTAKES

- Dividing mass by acceleration.
- Unit kg.

EXAMINER TIP

Show $F = ma$ before numbers.

QUESTION 59

Explain why a skydiver reaches terminal velocity.

QUESTION TYPE

Topic: Forces | **Command word:** explain | **Marks:** 3 | **Skill tested:** Terminal velocity

WHAT THE QUESTION IS ASKING

Link drag to increasing speed until forces balance.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

As speed increases, air resistance increases until it equals weight. Resultant force becomes zero so acceleration is zero and the skydiver falls at constant terminal velocity.

MARK BREAKDOWN

- Mark 1: Drag increases with speed.
- Mark 2: Forces balance at terminal velocity.
- Mark 3: Zero acceleration at constant speed.

COMMON MISTAKES

- Weight disappears.
- No mention of air resistance.

EXAMINER TIP

Terminal velocity when drag = weight.

QUESTION 60

Describe how force–extension data from a spring experiment can be used to find the spring constant.

QUESTION TYPE

Topic: Forces | **Command word:** describe | **Marks:** 3 | **Skill tested:** Hooke's law

WHAT THE QUESTION IS ASKING

Describe graph and gradient.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Measure force and extension for increasing loads within elastic limit. Plot force against extension. The spring constant k is the gradient of the linear region because $F = kx$.

MARK BREAKDOWN

- Mark 1: Measure F and x .
- Mark 2: Plot F against x .

Mark 3: k is gradient of linear region.

COMMON MISTAKES

- Plotting x against F without noting reciprocal.
- Using region beyond limit of proportionality.

EXAMINER TIP

Gradient of F – x graph gives k .

QUESTION 61

Compare elastic and inelastic deformation.

QUESTION TYPE

Topic: Forces | Command word: compare | Marks: 4 | Skill tested: Materials

WHAT THE QUESTION IS ASKING

Compare whether object returns to original shape.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Elastic deformation returns to original shape when force is removed, such as a spring within its limit. Inelastic deformation does not fully return, such as stretching a plastic ruler permanently.

MARK BREAKDOWN

- Mark 1: Elastic returns to original shape.
- Mark 2: Inelastic permanent change.
- Mark 3: Valid examples.
- Mark 4: Clear contrast.

COMMON MISTAKES

- Only one type described.
- Confusing with elastic potential energy only.

EXAMINER TIP

Elastic limit marks boundary on spring graphs.

QUESTION 62

A 0.40 kg ball travelling at 12 m/s hits a wall and rebounds at 8.0 m/s. The contact time is 0.020 s. Calculate the average force on the ball during the collision.

QUESTION TYPE

Topic: Forces | Command word: calculate | Marks: 4 | Skill tested: Momentum

WHAT THE QUESTION IS ASKING

Use $F = \Delta p / \Delta t$ with direction.

STEP-BY-STEP THINKING

1. Choose positive direction.
2. Calculate Δp including rebound.
3. Use $F = \Delta p / \Delta t$.
4. State unit N.

MODEL ANSWER

Take initial direction positive: $u = +12$, $v = -8$. $\Delta p = m(v-u) = 0.40 \times (-8-12) = -8.0 \text{ kg m/s}$. $F = \Delta p / \Delta t = -8.0 / 0.020 = -400 \text{ N}$. Magnitude 400 N (opposite to initial motion).

MARK BREAKDOWN

Mark 1: Momentum change includes direction.

Mark 2: $\Delta p = -8.0 \text{ kg m/s}$.

Mark 3: $F = \Delta p / \Delta t$ used.

Mark 4: Answer 400 N.

COMMON MISTAKES

- Ignoring rebound direction.
- Using speed only.

EXAMINER TIP

Rebound problems need signed velocities.

QUESTION 63

A lorry of mass 8000 kg slows from 25 m/s to 10 m/s in 5.0 s. Calculate the resultant force acting on the lorry.

QUESTION TYPE

Topic: Forces | Command word: calculate | Marks: 6 | Skill tested: Extended calculation

WHAT THE QUESTION IS ASKING

Use $F = ma$ after finding a , or $F = \Delta p / \Delta t$.

STEP-BY-STEP THINKING

1. Find acceleration $a = \Delta v / t$.
2. Use $F = ma$.
3. Substitute mass.
4. Include unit N.
5. Show working.
6. Correct sign/decision.

MODEL ANSWER

$a = (10 - 25) / 5.0 = -3.0 \text{ m/s}^2$. $F = 8000 \times (-3.0) = -24\,000 \text{ N}$. Retarding force 24 000 N.

MARK BREAKDOWN

Mark 1: $a = -3.0 \text{ m/s}^2$.

Mark 2: $F = ma$ used.

Mark 3: $m = 8000 \text{ kg}$.

Mark 4: Answer 24 000 N.

Mark 5: Unit N.

Mark 6: Working shown.

COMMON MISTAKES

- Using speed magnitudes only without deceleration sign.
- Dividing wrong way for a .

EXAMINER TIP

Deceleration: final minus initial over time.

QUESTION 64

Calculate the momentum of a 1500 kg car travelling at 20 m/s.

QUESTION TYPE

Topic: Forces | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Use $p = mv$.

STEP-BY-STEP THINKING

1. Write $p = mv$.
2. Multiply 1500×20 .
3. Unit kg m/s.

MODEL ANSWER

$p = 1500 \times 20 = 30\,000$ kg m/s.

MARK BREAKDOWN

- Mark 1: $p = mv$.
- Mark 2: Substitution.
- Mark 3: 30 000 kg m/s.

COMMON MISTAKES

- Adding mass and velocity.
- Unit N s only without equivalence note.

EXAMINER TIP

Momentum = mass $\times$ velocity.

QUESTION 65

Describe how you would investigate the relationship between force and extension for a spring up to its limit of proportionality.

QUESTION TYPE

Topic: Forces | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe apparatus and measurements.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Clamp spring vertically, attach known masses to find weight as force, measure extension with ruler from original length. Record force and extension, add masses incrementally, plot F against x , identify linear region.

MARK BREAKDOWN

- Mark 1: Force from known weights.
- Mark 2: Extension measured.
- Mark 3: F - x graph.
- Mark 4: Stay within limit of proportionality.

COMMON MISTAKES

- Measuring length not extension.
- No graph mention.

EXAMINER TIP

Extension = stretched length minus original length.

Section 2: Waves

QUESTION 66

State the equation linking wave speed, frequency, and wavelength.

QUESTION TYPE

Topic: Waves | Command word: state | Marks: 1 | Skill tested: Wave equation

WHAT THE QUESTION IS ASKING

Recall $v = f\lambda$.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Wave speed v = frequency $\times$ wavelength ($v = f\lambda$).

MARK BREAKDOWN

Mark 1: $v = f\lambda$ stated.

COMMON MISTAKES

- $v = f/\lambda$.
- Missing any quantity.

EXAMINER TIP

Say 'wave speed equals frequency times wavelength'.

QUESTION 67

Give the unit of frequency.

QUESTION TYPE

Topic: Waves | Command word: give | Marks: 1 | Skill tested: Knowledge recall

WHAT THE QUESTION IS ASKING

Name unit of frequency.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Hertz (Hz).

MARK BREAKDOWN

Mark 1: Hertz or Hz.

COMMON MISTAKES

- Metres per second.
- Seconds.

EXAMINER TIP

Hz = per second.

QUESTION 68

State what is meant by the amplitude of a wave.

QUESTION TYPE

Topic: Waves | Command word: state | Marks: 1 | Skill tested: Wave properties

WHAT THE QUESTION IS ASKING

Define amplitude.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Amplitude is the maximum displacement of a point on the wave from its rest position.

MARK BREAKDOWN

Mark 1: Maximum displacement from rest.

COMMON MISTAKES

- Whole wave length.
- Distance between peaks only without displacement.

EXAMINER TIP

Amplitude on graph is height from middle to peak.

QUESTION 69

Describe the difference between transverse and longitudinal waves.

QUESTION TYPE

Topic: Waves | Command word: describe | Marks: 2 | Skill tested: Wave types

WHAT THE QUESTION IS ASKING

Compare vibration direction to wave direction.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

In transverse waves, vibrations are perpendicular to wave direction, e.g. water waves. In longitudinal waves, vibrations are parallel to wave direction, e.g. sound.

MARK BREAKDOWN

Mark 1: Transverse: perpendicular vibrations with example.

Mark 2: Longitudinal: parallel vibrations with example.

COMMON MISTAKES

- Only one type.
- Wrong examples.

EXAMINER TIP

Light and water: transverse. Sound: longitudinal.

QUESTION 70

Explain why sound cannot travel through a vacuum.

QUESTION TYPE

Topic: Waves | Command word: explain | Marks: 2 | Skill tested: Sound

WHAT THE QUESTION IS ASKING

Link sound to particle vibrations.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Sound is a longitudinal wave that needs particles to vibrate and pass energy along. In a vacuum there are no particles to vibrate, so sound cannot propagate.

MARK BREAKDOWN

Mark 1: Sound needs medium/particles.

Mark 2: Vacuum has no particles.

COMMON MISTAKES

- Sound is blocked by vacuum 'walls'.
- Confusing with light.

EXAMINER TIP

Sound needs a medium; light does not.

QUESTION 71

A wave has frequency 500 Hz and wavelength 0.68 m. Calculate the wave speed.

QUESTION TYPE

Topic: Waves | Command word: calculate | Marks: 2 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Use $v = f\lambda$.

STEP-BY-STEP THINKING

1. Write $v = f\lambda$.
2. Multiply.
3. Unit m/s.

MODEL ANSWER

$v = 500 \times 0.68 = 340 \text{ m/s}$.

MARK BREAKDOWN

Mark 1: $v = f\lambda$.

Mark 2: 340 m/s.

COMMON MISTAKES

- Dividing.
- Unit Hz.

EXAMINER TIP

Show substitution before answer.

QUESTION 72

Explain how refraction occurs when light enters glass from air.

QUESTION TYPE

Topic: Waves | Command word: explain | Marks: 3 | Skill tested: Refraction

WHAT THE QUESTION IS ASKING

Link speed change to direction change.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Light slows down when entering denser glass so one part of the wavefront slows before the rest, causing the wave to change direction. The ray bends towards the normal.

MARK BREAKDOWN

- Mark 1: Speed decreases in glass.
- Mark 2: Wavefront change causes bending.
- Mark 3: Towards normal on entering denser medium.

COMMON MISTAKES

- Light speeds up entering glass.
- No wavefront explanation.

EXAMINER TIP

Refraction: speed change → direction change.

QUESTION 73

Describe how a ripple tank can be used to investigate reflection of water waves.

QUESTION TYPE

Topic: Waves | Command word: describe | Marks: 3 | Skill tested: Investigations

WHAT THE QUESTION IS ASKING

Describe apparatus and observations.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Generate straight ripples with a bar dipper toward a barrier at an angle. Observe incident and reflected wavefronts. Measure angles to normal for incident and reflected waves; they should be equal.

MARK BREAKDOWN

- Mark 1: Straight ripples toward barrier.
- Mark 2: Observe reflected wavefronts.

Mark 3: Angle of incidence equals angle of reflection.

COMMON MISTAKES

- No barrier.
- Confusing with refraction.

EXAMINER TIP

Ripple tanks show reflection and refraction visually.

QUESTION 74

Compare uses of ultraviolet and infrared radiation.

QUESTION TYPE

Topic: Waves | Command word: compare | Marks: 4 | Skill tested: EM spectrum

WHAT THE QUESTION IS ASKING

Compare applications of two EM regions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Ultraviolet is used in sterilisation and security marking because it causes fluorescence and kills microbes. Infrared is used in heating, thermal imaging, and remote controls because it transfers thermal energy and is detected by electronic sensors.

MARK BREAKDOWN

- Mark 1: Valid UV use.
Mark 2: Valid IR use.
Mark 3: Linked to properties.
Mark 4: Contrast between uses.

COMMON MISTAKES

- Same use for both.
- Only one region.

EXAMINER TIP

Link use to property: UV ionisation/fluorescence; IR heating.

QUESTION 75

A student stands 680 m from a cliff and hears an echo after 4.0 s. Calculate the speed of sound.

QUESTION TYPE

Topic: Waves | Command word: calculate | Marks: 4 | Skill tested: Echo calculation

WHAT THE QUESTION IS ASKING

Account for double distance.

STEP-BY-STEP THINKING

1. Distance travelled = 2×680 m.
2. Use $v = d/t$.
3. Unit m/s.

MODEL ANSWER

$d = 2 \times 680 = 1360 \text{ m}$. $v = 1360 \div 4.0 = 340 \text{ m/s}$.

MARK BREAKDOWN

- Mark 1: Double distance used.
- Mark 2: $v = d/t$.
- Mark 3: 340 m/s.
- Mark 4: Unit m/s.

COMMON MISTAKES

- Using 680 m only.
- Wrong time.

EXAMINER TIP

Echo: sound travels to cliff and back.

QUESTION 76

A television station sends the same programme through a cable and by radio wave at the same instant. Both signals travel 60 km to a house. Explain why the arrival times are almost identical, and suggest why the cable picture might appear slightly earlier on screen.

QUESTION TYPE

Topic: Waves | Command word: explain | Marks: 6 | Skill tested: EM waves

WHAT THE QUESTION IS ASKING

Compare travel times over the same distance and consider processing delays.

STEP-BY-STEP THINKING

1. Both signals are electromagnetic and travel at about the same speed over the same distance.
2. Use $t = d/v$ to show times are nearly equal for 60 km.
3. Cable feed may have less broadcasting/retransmission delay than the radio chain.
4. State that path length is the same for both in this scenario.
5. Use correct physics vocabulary.
6. Answer both parts clearly.

MODEL ANSWER

Both signals travel 60 km at approximately $3 \times 10^8 \text{ m/s}$, so $t = 60\,000 \div (3 \times 10^8) \approx 0.0002 \text{ s}$ for each — arrival times are almost identical. The cable picture may appear slightly earlier because the radio broadcast can include extra encoding and retransmission steps that add a small delay compared with the direct cable feed.

MARK BREAKDOWN

- Mark 1: Same distance and similar EM wave speed stated.
- Mark 2: Travel times nearly equal.
- Mark 3: Processing/retransmission delay for radio suggested.
- Mark 4: Clear logical structure.
- Mark 5: Correct terminology.
- Mark 6: Both parts addressed.

COMMON MISTAKES

- Saying lower frequency radio is slower over the same distance.
- Ignoring equal path length.

EXAMINER TIP

Same distance and speed → same travel time unless equipment adds delay.

QUESTION 77

Calculate the wavelength of a 2.4 MHz radio wave. Wave speed $c = 3.0 \times 10^8$ m/s.

QUESTION TYPE

Topic: Waves | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Rearrange $v = f\lambda$.

STEP-BY-STEP THINKING

1. Convert MHz to Hz.
2. Rearrange $\lambda = v/f$.
3. Calculate with unit m.

MODEL ANSWER

$f = 2.4 \times 10^6$ Hz. $\lambda = 3.0 \times 10^8 \div 2.4 \times 10^6 = 125$ m.

MARK BREAKDOWN

- Mark 1: Frequency converted.
 Mark 2: $\lambda = v/f$.
 Mark 3: 125 m.

COMMON MISTAKES

- Not converting MHz.
- Wrong power of ten.

EXAMINER TIP

Always convert prefix to base unit.

QUESTION 78

Describe how you would measure the speed of sound in air using two microphones connected to a data logger.

QUESTION TYPE

Topic: Waves | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe timing across known distance.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Place two microphones a measured distance apart facing a loud sharp sound source. Data logger records time difference between peaks. Speed = distance $\div$ time difference. Repeat and mean.

MARK BREAKDOWN

- Mark 1: Known separation measured.
 Mark 2: Time difference recorded.
 Mark 3: $v = d/t$.
 Mark 4: Repeat readings.

COMMON MISTAKES

- Single microphone.
- No distance measurement.

EXAMINER TIP

Sharp sound gives clear peak on trace.

Section 2: Magnetism and electromagnetism

QUESTION 79

State what is meant by a magnetic field.

QUESTION TYPE

Topic: Magnetism and electromagnetism | Command word: state | Marks: 1 | Skill tested: Magnetism

WHAT THE QUESTION IS ASKING

Define magnetic field.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A magnetic field is a region where a magnetic material or moving charge experiences a force.

MARK BREAKDOWN

Mark 1: Region where magnetic force acts.

COMMON MISTAKES

- Region with electric charge only.
- Area with gravity.

EXAMINER TIP

Field shown by iron filings or plotting compass.

QUESTION 80

Give the rule that links the direction of force on a current-carrying wire in a magnetic field.

QUESTION TYPE

Topic: Magnetism and electromagnetism | Command word: give | Marks: 1 | Skill tested: Motor effect

WHAT THE QUESTION IS ASKING

Name Fleming's left-hand rule.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fleming's left-hand rule.

MARK BREAKDOWN

Mark 1: Fleming's left-hand rule.

COMMON MISTAKES

- Right-hand rule for generators only without naming left for force.
- Ohm's law.

EXAMINER TIP

Left hand: force; right hand: induced current.

QUESTION 81

State what happens to the force on a wire in a magnetic field when the current is reversed.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** state | **Marks:** 1 | **Skill tested:** Motor effect

WHAT THE QUESTION IS ASKING

State effect of reversing current.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The direction of the force reverses.

MARK BREAKDOWN

Mark 1: Force direction reverses.

COMMON MISTAKES

- Force doubles.
- No change.

EXAMINER TIP

Reverse current or field → force reverses.

QUESTION 82

Describe the pattern and direction of magnetic field lines around a bar magnet.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** describe | **Marks:** 2 | **Skill tested:** Magnetic fields

WHAT THE QUESTION IS ASKING

Describe shape and direction.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Field lines leave the north pole and enter the south pole outside the magnet. Lines are closest where the field is strongest.

MARK BREAKDOWN

Mark 1: North to south outside magnet.
Mark 2: Density shows strength.

COMMON MISTAKES

- South to north outside.
- Lines cross.

EXAMINER TIP

Field lines never cross.

QUESTION 83

Explain how a simple electric motor produces rotation.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** explain | **Marks:** 2 | **Skill tested:** Motors

WHAT THE QUESTION IS ASKING

Link force on coil to split-ring commutator.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Current in the coil experiences forces due to the motor effect on opposite sides in opposite directions, producing a turning effect. The split-ring commutator reverses current every half turn so rotation continues in the same direction.

MARK BREAKDOWN

- Mark 1: Forces on coil create couple.
- Mark 2: Commutator reverses current each half turn.
- Mark 3: Continuous rotation.

COMMON MISTAKES

- No commutator role.
- Forces same direction on both sides.

EXAMINER TIP

Commutator keeps torque direction consistent.

QUESTION 84

Compare the function of a step-up transformer and a step-down transformer.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** compare | **Marks:** 2 | **Skill tested:** Transformers

WHAT THE QUESTION IS ASKING

Compare voltage changes.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A step-up transformer increases potential difference from primary to secondary; a step-down transformer decreases potential difference.

MARK BREAKDOWN

- Mark 1: Step-up increases voltage.

Mark 2: Step-down decreases voltage.

COMMON MISTAKES

- Both increase voltage.
- Transformers work on d.c.

EXAMINER TIP

Transformers need alternating current.

QUESTION 85

Explain electromagnetic induction in a wire moving through a magnetic field.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** explain | **Marks:** 3 | **Skill tested:** Induction

WHAT THE QUESTION IS ASKING

Explain how potential difference is induced.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

When a conductor cuts through magnetic field lines, a potential difference is induced across it. If the circuit is complete, an induced current flows. Faster cutting or stronger field increases induced p.d.

MARK BREAKDOWN

- Mark 1: Conductor cuts field lines.
 Mark 2: Induced p.d. across conductor.
 Mark 3: Current if complete circuit.

COMMON MISTAKES

- Magnetic field created by movement without cutting lines.
- Constant field linkage only at GCSE basic level confused.

EXAMINER TIP

Induction needs relative motion between conductor and field.

QUESTION 86

A transformer has 500 turns on the primary coil and 50 turns on the secondary. The primary p.d. is 230 V. Calculate the secondary p.d.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** calculate | **Marks:** 3 | **Skill tested:** Transformers

WHAT THE QUESTION IS ASKING

Use $V_p/V_s = N_p/N_s$.

STEP-BY-STEP THINKING

1. Write ratio equation.
2. Substitute.
3. Calculate V_s .

MODEL ANSWER

$V_s = V_p \times N_s/N_p = 230 \times 50/500 = 23 \text{ V}.$

MARK BREAKDOWN

Mark 1: Correct ratio.

Mark 2: Substitution.

Mark 3: 23 V.

COMMON MISTAKES

- Inverted ratio.
- Adding turns.

EXAMINER TIP

Step-down here: fewer secondary turns → lower voltage.

QUESTION 87

Describe how a generator produces an alternating potential difference.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** describe | **Marks:** 4 | **Skill tested:** Generators

WHAT THE QUESTION IS ASKING

Describe coil rotation in magnetic field.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Coil rotates in magnetic field so sides cut field lines. Induced p.d. reverses every half turn as direction of cutting reverses, producing a.c. Output voltage depends on speed of rotation and field strength.

MARK BREAKDOWN

Mark 1: Coil rotates cutting field lines.

Mark 2: Induced p.d. reverses each half turn.

Mark 3: a.c. output.

Mark 4: Factors affecting output.

COMMON MISTAKES

- Direct current from basic generator without split ring difference.
- No rotation.

EXAMINER TIP

Dynamos use slip rings for a.c.

QUESTION 88

Evaluate using renewable energy with wind turbines that contain electromagnets in their generators.

QUESTION TYPE

Topic: Magnetism and electromagnetism | **Command word:** evaluate | **Marks:** 4 | **Skill tested:** Applications

WHAT THE QUESTION IS ASKING

Weigh environmental and technical points.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.

2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Wind turbines generate a.c. using electromagnetic induction with low operating emissions, supporting low-carbon electricity. However, output is variable, turbines require maintenance, and grid infrastructure must handle fluctuating supply. Overall they reduce fossil fuel use where wind resource is good.

MARK BREAKDOWN

- Mark 1: Induction used to generate electricity.
- Mark 2: Low carbon advantage.
- Mark 3: Variability or maintenance limitation.
- Mark 4: Judgement.

COMMON MISTAKES

- No link to electromagnetism.
- One-sided.

EXAMINER TIP

Link physics principle to real energy debate.

QUESTION 89

A wire of length 0.35 m carries a current of 2.0 A at right angles to a magnetic field of flux density 0.40 T. Calculate the force on the wire.

QUESTION TYPE

Topic: Magnetism and electromagnetism | Command word: calculate | Marks: 6 | Skill tested: Motor effect calculation

WHAT THE QUESTION IS ASKING

Use $F = BIl$.

STEP-BY-STEP THINKING

1. Write $F = BIl$.
2. All quantities at right angles.
3. Multiply.
4. Unit N.
5. Show working.
6. State answer.

MODEL ANSWER

$$F = 0.40 \times 2.0 \times 0.35 = 0.28 \text{ N.}$$

MARK BREAKDOWN

- Mark 1: $F = BIl$.
- Mark 2: Substitution.
- Mark 3: 0.28 N.
- Mark 4: Unit N.
- Mark 5: Working.
- Mark 6: Right-angle condition noted.

COMMON MISTAKES

- Missing length.
- Wrong unit T only.

EXAMINER TIP

Force zero if wire parallel to field.

QUESTION 90

The primary coil of a transformer has 800 turns and the secondary has 40 turns. If the primary voltage is 240 V, calculate the secondary voltage.

QUESTION TYPE

Topic: Magnetism and electromagnetism | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Use turns ratio.

STEP-BY-STEP THINKING

1. $V_s/V_p = N_s/N_p$.
2. Substitute.
3. Answer in V.

MODEL ANSWER

$V_s = 240 \times 40/800 = 12 \text{ V}$.

MARK BREAKDOWN

- Mark 1: Correct ratio.
- Mark 2: Substitution.
- Mark 3: 12 V.

COMMON MISTAKES

- Multiplying by 800/40 incorrectly inverted.

EXAMINER TIP

Fewer secondary turns → step-down.

QUESTION 91

Describe how you would investigate the force on a current-carrying wire in a magnetic field using a top pan balance and wire on a pivot.

QUESTION TYPE

Topic: Magnetism and electromagnetism | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe measuring force change when current switched on.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Place magnet on balance with wire between poles horizontal at right angles. Zero balance with current off. Switch on current, record balance reading change equal to force on wire. Repeat for different currents, plot force against current.

MARK BREAKDOWN

- Mark 1: Wire perpendicular to field.
- Mark 2: Balance measures force change.
- Mark 3: Vary current.
- Mark 4: Repeat and graph.

COMMON MISTAKES

- Wire parallel to field.
- No repeat readings.

EXAMINER TIP

Motor effect practical: F increases with current.

Section 2: Space physics

QUESTION 92

State what provides the centripetal force on a planet orbiting the Sun.

QUESTION TYPE

Topic: Space physics | Command word: state | Marks: 1 | Skill tested: Orbits

WHAT THE QUESTION IS ASKING

Identify force keeping planet in orbit.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Gravitational force (gravity) from the Sun.

MARK BREAKDOWN

Mark 1: Gravitational force / gravity.

COMMON MISTAKES

- Magnetic force.
- Air resistance.

EXAMINER TIP

Orbits: gravity provides centripetal force.

QUESTION 93

Give the name of our galaxy.

QUESTION TYPE

Topic: Space physics | Command word: give | Marks: 1 | Skill tested: Astronomy

WHAT THE QUESTION IS ASKING

Name the Milky Way.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The Milky Way.

MARK BREAKDOWN

Mark 1: Milky Way.

COMMON MISTAKES

- Andromeda.
- Solar System.

EXAMINER TIP

Solar System is within the Milky Way.

QUESTION 94

State what is meant by a light-year.

QUESTION TYPE

Topic: Space physics | **Command word:** state | **Marks:** 1 | **Skill tested:** Distances

WHAT THE QUESTION IS ASKING

Define light-year.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A light-year is the distance light travels in one year in a vacuum.

MARK BREAKDOWN

Mark 1: Distance light travels in one year.

COMMON MISTAKES

- Time light takes to travel.
- Unit of speed.

EXAMINER TIP

Light-year is distance, not time.

QUESTION 95

Describe the life cycle of a star similar in mass to the Sun.

QUESTION TYPE

Topic: Space physics | **Command word:** describe | **Marks:** 2 | **Skill tested:** Stellar life cycle

WHAT THE QUESTION IS ASKING

Outline main stages.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Nebula collapses to form a main sequence star where hydrogen fuses to helium. Later it becomes a red giant, then sheds outer layers forming a planetary nebula, leaving a white dwarf.

MARK BREAKDOWN

- Mark 1: Nebula to main sequence.
- Mark 2: Red giant stage.
- Mark 3: White dwarf end point.

COMMON MISTAKES

- Supernova for Sun-like star.
- Black hole ending.

EXAMINER TIP

Sun-like stars do not become supernovae.

QUESTION 96

Explain why fusion in stars releases energy.

QUESTION TYPE

Topic: Space physics | Command word: explain | Marks: 2 | Skill tested: Fusion

WHAT THE QUESTION IS ASKING

Link mass defect to energy release.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Fusion combines light nuclei; the mass of the product nucleus is slightly less than the sum of original nuclei. The mass difference is converted to energy according to $E = mc^2$.

MARK BREAKDOWN

- Mark 1: Light nuclei fuse.
Mark 2: Mass decreases.
Mark 3: Energy released via $E = mc^2$.

COMMON MISTAKES

- Energy created without mass change.
- Fission described instead.

EXAMINER TIP

Fusion needs very high temperature and pressure.

QUESTION 97

Compare geostationary and low Earth orbit satellites.

QUESTION TYPE

Topic: Space physics | Command word: compare | Marks: 2 | Skill tested: Satellites

WHAT THE QUESTION IS ASKING

Compare orbit height and period uses.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Geostationary orbit is at about 36 000 km above the equator with orbital period 24 hours, used for communications. Low Earth orbit is much closer (hundreds of km) with shorter period, used for imaging and ISS.

MARK BREAKDOWN

Mark 1: Geostationary period 24 h / comms.

Mark 2: LEO closer / shorter period / imaging.

Mark 3: Both valid differences.

COMMON MISTAKES

- Same height.
- Only one orbit described.

EXAMINER TIP

Geostationary: fixed above equator.

QUESTION 98

Explain why the Universe is expanding using evidence from distant galaxies.

QUESTION TYPE

Topic: Space physics | Command word: explain | Marks: 3 | Skill tested: Red-shift

WHAT THE QUESTION IS ASKING

Link red-shift to recession.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Light from distant galaxies shows red-shift, meaning wavelengths are stretched as galaxies move away from us. More distant galaxies have greater red-shift, showing faster recession and supporting expansion of the Universe.

MARK BREAKDOWN

Mark 1: Red-shift observed.

Mark 2: Galaxies moving away.

Mark 3: More distant → greater red-shift → faster recession.

COMMON MISTAKES

- Blue-shift for all.
- Stars not galaxies.

EXAMINER TIP

Hubble's law: recession speed increases with distance.

QUESTION 99

Describe how the change in wavelength of light from a galaxy can be measured.

QUESTION TYPE

Topic: Space physics | Command word: describe | Marks: 3 | Skill tested: Red-shift measurement

WHAT THE QUESTION IS ASKING

Describe spectrum comparison.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.

4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Split galaxy light into a spectrum using a diffraction grating or prism. Compare observed absorption lines with known laboratory wavelengths. The shift toward longer wavelength (red end) indicates red-shift.

MARK BREAKDOWN

- Mark 1: Spectrum obtained.
- Mark 2: Known line positions compared.
- Mark 3: Shift to longer wavelength measured.

COMMON MISTAKES

- Measuring brightness only.
- No reference to spectral lines.

EXAMINER TIP

Spectral line patterns act as fingerprints.

QUESTION 100

Evaluate using space telescopes compared with ground-based telescopes.

QUESTION TYPE

Topic: Space physics | Command word: evaluate | Marks: 4 | Skill tested: Observing universe

WHAT THE QUESTION IS ASKING

Weigh advantages and limitations.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Space telescopes avoid atmospheric absorption and turbulence, observing clearer infrared/UV that does not reach ground. They are expensive, hard to maintain, and have limited lifetime. Ground telescopes are cheaper and easier to upgrade but affected by weather and atmosphere.

MARK BREAKDOWN

- Mark 1: Space: no atmosphere distortion.
- Mark 2: Space: broader EM range.
- Mark 3: Ground: cheaper/easier access.
- Mark 4: Overall judgement.

COMMON MISTAKES

- Space telescopes always cheaper.
- No atmospheric issue mentioned.

EXAMINER TIP

Hubble and JWST examples of space advantage.

QUESTION 101

A galaxy recedes at 2.4×10^8 m/s. The speed of light is 3.0×10^8 m/s. Calculate the red-shift z if defined as $\Delta\lambda/\lambda = v/c$ for non-relativistic speeds.

QUESTION TYPE

Topic: Space physics | Command word: calculate | Marks: 4 | Skill tested: Red-shift calculation

WHAT THE QUESTION IS ASKING

Use $z = v/c$.

STEP-BY-STEP THINKING

1. Write $z = v/c$.
2. Substitute.
3. No unit for z .

MODEL ANSWER

$$z = 2.4 \times 10^8 \div 3.0 \times 10^8 = 0.008.$$

MARK BREAKDOWN

- Mark 1: $z = v/c$.
- Mark 2: Substitution.
- Mark 3: 0.008.
- Mark 4: Correct decimal.

COMMON MISTAKES

- Multiplying v and c .
- Unit m/s on z .

EXAMINER TIP

Red-shift z is a ratio — no unit.

QUESTION 102

Describe how observations of supernovae in distant galaxies provide evidence for the accelerating expansion of the Universe.

QUESTION TYPE

Topic: Space physics | Command word: describe | Marks: 6 | Skill tested: Cosmology

WHAT THE QUESTION IS ASKING

Link brightness distance to expansion rate change.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Type Ia supernovae have known intrinsic brightness so distance can be estimated. Observed brightness compared to red-shift distance shows galaxies farther than expected for a given red-shift, implying expansion is accelerating. Dark energy is suggested to explain this.

MARK BREAKDOWN

- Mark 1: Supernova standard candle idea.
- Mark 2: Distance from brightness.
- Mark 3: Compared with red-shift.
- Mark 4: Expansion accelerating conclusion.

Mark 5: Dark energy mention optional extension.

Mark 6: Coherent account.

COMMON MISTAKES

- Supernova causes expansion.
- No link to brightness and distance.

EXAMINER TIP

Standard candles allow distance comparison.

QUESTION 103

Light from the Sun takes 8.3 minutes to reach Earth. Speed of light is 3.0×10^8 m/s. Calculate the distance from the Sun to Earth.

QUESTION TYPE

Topic: Space physics | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Use $d = vt$ with time in seconds.

STEP-BY-STEP THINKING

1. Convert 8.3 min to seconds.
2. $d = vt$.
3. Answer in m or km.

MODEL ANSWER

$t = 8.3 \times 60 = 498$ s. $d = 3.0 \times 10^8 \times 498 = 1.49 \times 10^{11}$ m (about 150 million km).

MARK BREAKDOWN

- Mark 1: Time in seconds.
 Mark 2: $d = vt$.
 Mark 3: Correct order of magnitude.

COMMON MISTAKES

- Using minutes directly.
- Wrong speed.

EXAMINER TIP

Astronomy distances often given in metres or km.

QUESTION 104

Describe how you would use published orbital data and the equation speed = distance ÷ time to estimate the radius of a geostationary satellite orbit.

QUESTION TYPE

Topic: Space physics | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe data sources and calculation steps for an orbital estimate.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Look up the orbital period of a geostationary satellite (24 hours) and its orbital speed from a reliable data source. Convert period to seconds, calculate orbital circumference using $\text{speed} \times \text{time}$, then radius using $\text{circumference} = 2\pi r$. Compare with textbook value to assess accuracy.

MARK BREAKDOWN

Mark 1: Orbital period and speed obtained from data.

Mark 2: Circumference calculated from $\text{speed} \times \text{time}$.

Mark 3: Radius found from $C = 2\pi r$.

Mark 4: Comparison or repeat/check mentioned.

COMMON MISTAKES

- Using period in hours without conversion.
- Confusing radius with diameter.

EXAMINER TIP

Geostationary orbit period matches Earth's rotation: about 24 hours.

Section 2: Required practical skills

QUESTION 105

State what is meant by an independent variable.

QUESTION TYPE

Topic: Required practical skills | Command word: state | Marks: 1 | Skill tested: Working scientifically

WHAT THE QUESTION IS ASKING

Define independent variable.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The independent variable is the one deliberately changed by the experimenter.

MARK BREAKDOWN

Mark 1: Variable changed deliberately.

COMMON MISTAKES

- Variable measured.
- Variable kept constant.

EXAMINER TIP

Independent = you change it.

QUESTION 106

Give one hazard associated with using a mains immersion heater in a density or SHC practical.

QUESTION TYPE

Topic: Required practical skills | Command word: give | Marks: 1 | Skill tested: Safety

WHAT THE QUESTION IS ASKING

Name a hazard.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Risk of electric shock or burns from hot water/heater.

MARK BREAKDOWN

Mark 1: Electric shock or burns/scalds.

COMMON MISTAKES

- Paper cut.
- Radiation poisoning from heater.

EXAMINER TIP

Link hazard to control: RCD, keep dry.

QUESTION 107

State why repeat readings are taken in physics investigations.

QUESTION TYPE

Topic: Required practical skills | Command word: state | Marks: 1 | Skill tested: Data quality

WHAT THE QUESTION IS ASKING

State purpose of repeats.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

To identify anomalies and calculate a mean, improving reliability.

MARK BREAKDOWN

Mark 1: Identify anomalies and/or calculate mean.

Mark 2: Improve reliability.

COMMON MISTAKES

- To change independent variable.
- To make results bigger.

EXAMINER TIP

Reliability improves with repeats and means.

QUESTION 108

Describe how to identify an anomalous result in a set of repeat measurements.

QUESTION TYPE

Topic: Required practical skills | Command word: describe | Marks: 2 | Skill tested: Data analysis

WHAT THE QUESTION IS ASKING

Describe comparison with other repeats.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Compare each repeat to others. A result that is very different from the others at the same conditions is anomalous and should be excluded before calculating the mean.

MARK BREAKDOWN

Mark 1: Compare repeats.

Mark 2: Very different value identified as anomaly.

Mark 3: Exclude before mean.

COMMON MISTAKES

- Remove highest value always.
- Never exclude data.

EXAMINER TIP

Do not delete without justification.

QUESTION 109

Explain why a line of best fit is drawn for linear relationships.

QUESTION TYPE

Topic: Required practical skills | Command word: explain | Marks: 2 | Skill tested: Graphs

WHAT THE QUESTION IS ASKING

Explain averaging scatter.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Experimental points include random error scatter. A line of best fit shows the overall trend by balancing points above and below the line, giving a more reliable relationship than joining dot-to-dot.

MARK BREAKDOWN

Mark 1: Shows overall trend.

Mark 2: Accounts for scatter/random error.

Mark 3: More reliable than dot-to-dot.

COMMON MISTAKES

- Joins every point exactly.
- Removes systematic error automatically.

EXAMINER TIP

Line of best fit: roughly equal points above/below.

QUESTION 110

Compare systematic errors and random errors.

QUESTION TYPE

Topic: Required practical skills | Command word: compare | Marks: 2 | Skill tested: Errors

WHAT THE QUESTION IS ASKING

Compare causes and effects.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Random errors cause scatter around the true value and are reduced by repeats and mean. Systematic errors shift all readings consistently, such as zero error on a balance, and are not removed by averaging.

MARK BREAKDOWN

Mark 1: Random: scatter, reduced by repeats.

Mark 2: Systematic: consistent shift, e.g. zero error.

Mark 3: Contrast clear.

COMMON MISTAKES

- Same definition for both.
- Only random described.

EXAMINER TIP

Systematic affects accuracy; random affects precision.

QUESTION 111

Describe how to calculate a mean value from five repeat readings.

QUESTION TYPE

Topic: Required practical skills | **Command word:** describe | **Marks:** 3 | **Skill tested:** Processing data

WHAT THE QUESTION IS ASKING

Describe summing and dividing.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Add the five values together and divide by five. If an anomaly was identified, exclude it first, note reduced number of readings, and divide by the number remaining.

MARK BREAKDOWN

Mark 1: Sum values.

Mark 2: Divide by number of readings.

Mark 3: Anomaly exclusion if appropriate.

COMMON MISTAKES

- Multiply readings.
- Divide by highest.

EXAMINER TIP

Mean = total $\div$ number.

QUESTION 112

Explain why calibration improves measurement accuracy.

QUESTION TYPE

Topic: Required practical skills | **Command word:** explain | **Marks:** 3 | **Skill tested:** Equipment

WHAT THE QUESTION IS ASKING

Link calibration to true values.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.

2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Calibration compares instrument readings to known standards and adjusts for systematic error so readings match true values more closely, improving accuracy.

MARK BREAKDOWN

- Mark 1: Compares to known standard.
- Mark 2: Corrects systematic error.
- Mark 3: Improves accuracy.

COMMON MISTAKES

- Increases random scatter.
- Removes need for repeats.

EXAMINER TIP

Accuracy = closeness to true value.

QUESTION 113

Evaluate using digital data loggers instead of manual stopwatches for timing oscillations.

QUESTION TYPE

Topic: Required practical skills | Command word: evaluate | Marks: 4 | Skill tested: Equipment choice

WHAT THE QUESTION IS ASKING

Weigh precision and practical issues.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Data loggers record at high frequency with less reaction time error, improving precision for fast events. However, they cost more, need setup, and may introduce software delays if poorly configured. Overall better for short or rapid events.

MARK BREAKDOWN

- Mark 1: Logger reduces reaction time error.
- Mark 2: Higher sampling rate.
- Mark 3: Cost/setup limitation.
- Mark 4: Judgement for oscillations.

COMMON MISTAKES

- Stopwatch always better.
- No reaction time issue mentioned.

EXAMINER TIP

Reaction time ~ 0.2 s limits stopwatches.

QUESTION 114

Describe how to estimate uncertainty when measuring the length of a wire with a metre rule to the nearest millimetre.

QUESTION TYPE

Topic: Required practical skills | **Command word:** describe | **Marks:** 4 | **Skill tested:** Uncertainty

WHAT THE QUESTION IS ASKING

Describe half smallest division.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The smallest division is 1 mm so each reading has uncertainty of ± 0.5 mm. For a single measurement report length ± 0.5 mm; for a mean from repeats, uncertainty can be reduced but instrument uncertainty remains at least ± 0.5 mm unless finer instrument used.

MARK BREAKDOWN

- Mark 1: $\pm$ half smallest division.
- Mark 2: 1 mm scale $\rightarrow \pm 0.5$ mm.
- Mark 3: Apply to wire length.
- Mark 4: Link to reporting.

COMMON MISTAKES

- ± 1 mm always without justification.
- No reference to scale.

EXAMINER TIP

Resolution determines minimum uncertainty.

QUESTION 115

Plan an investigation to test whether the period of a pendulum depends on its mass. Describe independent variable, dependent variable, controls, equipment, and expected graph.

QUESTION TYPE

Topic: Required practical skills | **Command word:** describe | **Marks:** 6 | **Skill tested:** Planning investigations

WHAT THE QUESTION IS ASKING

Full plan with variables and method.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Independent variable: mass of bob. Dependent variable: period T measured for 10 oscillations divided by 10. Control: length, amplitude ($< 10^\circ$), same gravitational field. Equipment: clamp stand, string, masses, stopwatch/ruler. Expect period independent of mass — horizontal trend on T vs mass graph within scatter.

MARK BREAKDOWN

Mark 1: IV mass, DV period.

Mark 2: Controls: length, amplitude.

Mark 3: Method for measuring T.

Mark 4: Equipment listed.

Mark 5: Expected graph/trend.

Mark 6: Coherent plan.

COMMON MISTAKES

- Length as IV when question asks mass.
- No controls.

EXAMINER TIP

Period formula $T = 2\pi\sqrt{l/g}$ — no mass term.

QUESTION 116

A student measures wire length five times (cm): 98.2, 98.5, 98.4, 98.3, 98.4. Calculate the mean length.

QUESTION TYPE

Topic: Required practical skills | Command word: calculate | Marks: 3 | Skill tested: Calculation

WHAT THE QUESTION IS ASKING

Sum and divide.

STEP-BY-STEP THINKING

1. Add five values.
2. Divide by 5.
3. Unit cm.

MODEL ANSWER

Mean = $(98.2+98.5+98.4+98.3+98.4)/5 = 491.8/5 = 98.4$ cm.

MARK BREAKDOWN

- Mark 1: Sum correct.
 Mark 2: Divide by 5.
 Mark 3: 98.4 cm.

COMMON MISTAKES

- Using median without instruction.
- Wrong denominator.

EXAMINER TIP

Show addition then division in working.

QUESTION 117

Describe how you would safely investigate the resistance of a thermistor at different temperatures using a water bath, thermometer, and circuit.

QUESTION TYPE

Topic: Required practical skills | Command word: describe | Marks: 4 | Skill tested: Required practical skills

WHAT THE QUESTION IS ASKING

Describe varying temperature and measuring R.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.

3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Connect thermistor in series with ammeter and supply, voltmeter in parallel. Place in water bath with thermometer. Heat bath, record temperature, current, and p.d. at intervals, calculate $R = V/I$ each time. Keep other variables constant, plot R against temperature.

MARK BREAKDOWN

- Mark 1: $R = V/I$ at each temperature.
- Mark 2: Water bath and thermometer.
- Mark 3: Safety: dry hands, low voltage, support beaker.
- Mark 4: Graph R vs T .

COMMON MISTAKES

- No thermometer.
- Measuring resistance without V and I .

EXAMINER TIP

Thermistor R decreases with temperature for NTC type — note in conclusion.

Section 3: Calculation walkthroughs

Calculation 1: Gravitational potential energy

A 3.5 kg mass is lifted through 2.0 m. Calculate the change in gravitational potential energy. Use $g = 9.8 \text{ N/kg}$.

Formula: $E_p = mgh$

Rearrangement: E_p is already subject of formula

Substitution: $E_p = 3.5 \times 9.8 \times 2.0$

Working: $E_p = 68.6 \text{ J}$

Answer: 69 J (2 s.f.) or 68.6 J

Use mass in kg, height in m, g in N/kg.

Calculation 2: Kinetic energy

Calculate the kinetic energy of a 1200 kg car travelling at 15 m/s.

Formula: $E_k = \frac{1}{2}mv^2$

Rearrangement: E_k is subject

Substitution: $E_k = 0.5 \times 1200 \times 15^2$

Working: $E_k = 0.5 \times 1200 \times 225 = 135\,000 \text{ J}$

Answer: 135 000 J or $1.35 \times 10^5 \text{ J}$

Remember to square the speed only.

Calculation 3: Power

A kettle transfers 180 000 J in 60 s. Calculate its power.

Formula: $P = E/t$

Rearrangement: P is subject

Substitution: $P = 180\,000 \div 60$

Working: $P = 3000 \text{ W}$

Answer: 3000 W or 3.0 kW

Power in watts when energy is in joules and time in seconds.

Calculation 4: Charge flow

A current of 0.25 A flows for 4 minutes. Calculate the charge flow.

Formula: $Q = It$

Rearrangement: Q is subject

Substitution: $t = 240 \text{ s}$; $Q = 0.25 \times 240$

Working: $Q = 60 \text{ C}$

Answer: 60 C

Convert minutes to seconds before substituting.

Calculation 5: Potential difference and resistance

A resistor of $25 \, \Omega$ has a current of 0.40 A. Calculate the potential difference across it.

Formula: $V = IR$

Rearrangement: V is subject

Substitution: $V = 0.40 \times 25$

Working: $V = 10 \text{ V}$

Answer: 10 V

Ohm's law: V , I , R must be in volts, amps, ohms.

Calculation 6: Density

An object has mass 2.4 kg and volume 0.0030 m^3 . Calculate its density.

Formula: $\rho = m/V$

Rearrangement: ρ is subject

Substitution: $\rho = 2.4 \div 0.0030$

Working: $\rho = 800 \text{ kg/m}^3$

Answer: 800 kg/m^3

Density connects mass and volume — check units match.

Calculation 7: Specific heat capacity

How much energy is needed to heat 0.50 kg of water from 20°C to 50°C ? Specific heat capacity of water = $4200 \text{ J/kg }^\circ\text{C}$.

Formula: $E = mc\Delta T$

Rearrangement: E is subject; $\Delta T = 30^\circ\text{C}$

Substitution: $E = 0.50 \times 4200 \times 30$

Working: $E = 63\,000 \text{ J}$

Answer: $63\,000 \text{ J}$ or $6.3 \times 10^4 \text{ J}$

Use temperature change, not final temperature alone.

Calculation 8: Resultant force and acceleration

Calculate the acceleration of a 950 kg car when the resultant force is 1900 N .

Formula: $F = ma$

Rearrangement: $a = F/m$

Substitution: $a = 1900 \div 950$

Working: $a = 2.0 \text{ m/s}^2$

Answer: 2.0 m/s^2

Resultant force determines acceleration.

Calculation 9: Momentum

Calculate the momentum of a 0.20 kg ball moving at 25 m/s .

Formula: $p = mv$

Rearrangement: p is subject

Substitution: $p = 0.20 \times 25$

Working: $p = 5.0 \text{ kg m/s}$

Answer: 5.0 kg m/s

Momentum is a vector; include direction in full problems.

Calculation 10: Wave speed

A wave has frequency 50 Hz and wavelength 6.0 m. Calculate wave speed.

Formula: $v = f\lambda$

Rearrangement: v is subject

Substitution: $v = 50 \times 6.0$

Working: $v = 300 \text{ m/s}$

Answer: 300 m/s

Applies to all waves including electromagnetic waves.

Calculation 11: Transformer turns ratio

A transformer has 1200 primary turns and 120 secondary turns. The primary p.d. is 240 V. Calculate the secondary p.d.

Formula: $V_s/V_p = N_s/N_p$

Rearrangement: $V_s = V_p \times N_s/N_p$

Substitution: $V_s = 240 \times 120/1200$

Working: $V_s = 24 \text{ V}$

Answer: 24 V

Step-down transformer: fewer secondary turns $\rightarrow$ lower voltage.

Section 4: Required practical walkthroughs

Practical 1: Investigating the resistance of a wire

Aim: To determine how the resistance of a wire depends on its length.

Variables: Independent: length of wire. Dependent: resistance (calculated from V and I). Control: material, diameter, temperature.

Method: Connect a length of constantan wire in series with ammeter, switch, and low-voltage supply. Connect voltmeter in parallel with the wire. Measure length with metre rule. Record I and V , calculate $R = V/I$. Repeat for different lengths with same thickness and material.

Risk assessment: Wire may become hot if current too high — use low voltage and short bursts. Ensure connections secure to prevent short circuit. Eye protection if tensioned wire could snap.

Results table: Table columns: length/m, current/A, potential difference/V, resistance/ Ω . Take repeat R for each length and calculate mean.

Graph guidance: Plot mean resistance (y) against length (x). Expect straight line through origin if R proportional to length.

Conclusion: Resistance increases proportionally with length because longer wire provides more resistance to charge flow.

Evaluation: Random error from reading metres and meters — use repeats. Systematic error if contact resistance at crocodile clips — minimise by same clip positions. Improve by using thicker wire to reduce heating or use data logger.

Common exam questions:

- Why must only length be changed?
- How is resistance calculated from measurements?
- What graph shape shows $R \propto L$?

Model answers:

- So the relationship between length and resistance is isolated; other factors affect R .
- Measure V and I , then $R = V/I$ for each length.
- Straight line through origin on R vs length graph.

Practical 2: Measuring the density of a regular solid

Aim: To measure the density of a regular wooden block using a balance and ruler.

Variables: Independent: object (block). Dependent: density $\rho = m/V$. Measure mass and dimensions.

Method: Measure mass on balance. Measure length, width, height with ruler at several points, calculate mean dimensions. Volume $V = lwh$. Calculate $\rho = m/V$. Repeat entire procedure.

Risk assessment: Ruler edges may slip — clamp block if needed. Balance is precision equipment — avoid spills.

Results table: Record mass/kg, length/m, width/m, height/m, volume/ m^3 , density/ $kg\ m^{-3}$. Include repeats and mean.

Graph guidance: Optional bar chart comparing mean density to textbook value for material.

Conclusion: Compare calculated density to known value for wood ($\sim 500\text{--}800\ kg/m^3$ depending on type). Percentage difference estimates accuracy.

Evaluation: Uncertainty in ruler $\pm 0.5\ mm$ affects volume. Improve using vernier calipers/micrometer. Irregular objects need displacement can method.

Common exam questions:

- Why measure dimensions at several places?

- How would you find density of an irregular stone?
- Why include units in the density answer?

Model answers:

- Block may not be perfectly uniform; mean dimensions improve reliability.
- Use displacement can/eureka can to find volume, mass on balance, $\rho = m/V$.
- Density has units (kg/m^3) — values meaningless without them.

Section 5: Full mini mock paper

Total marks: 50

Question 1 (1 marks)

State the unit of power.

Question 2 (1 marks)

Give one non-renewable energy resource.

Question 3 (4 marks)

Describe the difference between series and parallel circuits.

Question 4 (2 marks)

Explain why metals conduct electricity.

Question 5 (2 marks)

A $5.0\ \Omega$ resistor is connected to 10 V. Calculate the current.

Question 6 (3 marks)

Explain why insulating a loft reduces energy wasted from a house.

Question 7 (3 marks)

Describe the particle arrangement in a gas.

Question 8 (4 marks)

Compare alpha and gamma radiation in terms of ionisation and penetration.

Question 9 (4 marks)

A 2.0 kg object falls at terminal velocity. Explain the forces acting.

Question 10 (3 marks)

Calculate the wave speed of a 40 Hz wave with wavelength 8.5 m.

Question 11 (4 marks)

Describe how a step-up transformer helps reduce energy loss in National Grid cables.

Question 12 (6 marks)

Evaluate the use of nuclear power for generating electricity.

Question 13 (3 marks)

A 800 W kettle is used for 3 minutes. Calculate energy transferred.

Question 14 (4 marks)

Describe how you would measure the density of an irregular object.

Question 15 (6 marks)

A car of mass 1000 kg changes velocity from 10 m/s to 25 m/s in 5.0 s. Calculate resultant force.

Mini mock — step-by-step walkthrough answers

Answer 1

Power is measured in watts (W).

Mark breakdown:

Mark 1: Watt or W.

Answer 2

Any fossil fuel or nuclear, e.g. natural gas or uranium fuel.

Mark breakdown:

Mark 1: Valid non-renewable named.

Answer 3

In series, components share one path so current is the same everywhere and potential differences add up. In parallel, each branch has the same p.d. across it and the total current splits between branches.

Mark breakdown:

Mark 1: Series: one path / same current.

Mark 2: Series: p.d.s shared/add.

Mark 3: Parallel: same p.d. per branch.

Mark 4: Parallel: current splits.

Answer 4

Metals contain delocalised electrons that move when a p.d. is applied, carrying charge as electric current.

Mark breakdown:

Mark 1: Delocalised electrons.

Mark 2: Move to carry charge/current.

Answer 5

$I = V/R = 10/5.0 = 2.0 \text{ A}$.

Mark breakdown:

Mark 1: $V = IR$ rearranged.

Mark 2: 2.0 A with unit.

Answer 6

Insulation reduces thermal energy transfer through the roof by conduction/convection/radiation. Less energy leaves so less fuel/electricity is needed to replace it.

Mark breakdown:

Mark 1: Reduces energy transfer out.

Mark 2: Less energy input needed.

Mark 3: Lower waste/cost linked.

Answer 7

Gas particles are far apart, move randomly at high speed, and collide with container walls.

Mark breakdown:

Mark 1: Far apart.

Mark 2: Random motion.

Mark 3: Collisions with walls.

Answer 8

Alpha is highly ionising but stopped by paper/skin; gamma is weakly ionising but penetrates thick lead. Alpha dangerous if ingested; gamma external hazard.

Mark breakdown:

Mark 1: Alpha ionisation high.

Mark 2: Alpha penetration low.

Mark 3: Gamma ionisation lower.

Mark 4: Gamma penetration high.

Answer 9

Weight acts downward. Air resistance increases with speed until equals weight. Resultant force zero so velocity constant.

Mark breakdown:

Mark 1: Weight downward.

Mark 2: Drag increases.

Mark 3: Forces balance.

Mark 4: Constant velocity.

Answer 10

$$v = f\lambda = 40 \times 8.5 = 340 \text{ m/s.}$$

Mark breakdown:

Mark 1: $v = f\lambda$.

Mark 2: Substitution.

Mark 3: 340 m/s.

Answer 11

Step-up increases voltage so current is lower for same power. Lower current reduces I^2R heating losses in cables.

Mark breakdown:

Mark 1: Increases voltage.

Mark 2: Lower current.

Mark 3: Less I^2R loss.

Mark 4: Clear link.

Answer 12

Fission gives large energy and low CO_2 in operation but produces radioactive waste and accident risk. Reliable baseload. Overall low-carbon if waste managed.

Mark breakdown:

Mark 1: Advantage: energy/low CO_2 .

Mark 2: Disadvantage: waste/risk.

Mark 3: Reliability.

Mark 4: Further balanced point.

Mark 5: Judgement.

Mark 6: Coherent evaluation.

Answer 13

$t = 180 \text{ s}$. $E = Pt = 800 \times 180 = 144\,000 \text{ J}$.

Mark breakdown:

Mark 1: Time in seconds.

Mark 2: $E = Pt$.

Mark 3: 144 000 J.

Answer 14

Mass on balance. Volume by measuring cylinder/displacement can. $\rho = m/V$ with repeats and mean.

Mark breakdown:

Mark 1: Mass measured.

Mark 2: Volume by displacement.

Mark 3: $\rho = m/V$.

Mark 4: Repeats/mean.

Answer 15

$a = (25-10)/5.0 = 3.0 \text{ m/s}^2$. $F = ma = 1000 \times 3.0 = 3000 \text{ N}$.

Mark breakdown:

Mark 1: $a = 3.0 \text{ m/s}^2$.

Mark 2: $F = ma$.

Mark 3: m correct.

Mark 4: 3000 N.

Mark 5: Unit N.

Mark 6: Working shown.